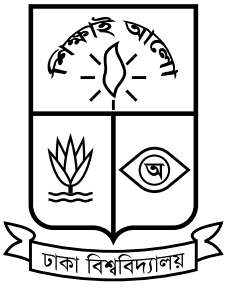
# UNIVERSITY OF DHAKA

Department of Computer Science and Engineering




# AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization

**Initial Thesis Progress Report**

Application Scope: Bangladeshi SME Businesses

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# Declaration Form

This progress report presents the research foundation, current implementation, and development-stage experimental evidence for the thesis project entitled *AI-Powered Business Analytics System for Performance Evaluation, Insight Extraction, and Strategy Optimization*. The work is conducted by the authors under the supervision of Dr. Saifuddin Md. Tareeq. Ideas, methods, datasets, software libraries, and findings originating from prior work are acknowledged and cited. The numerical forecasting results are restricted to the supplied daily shop–category panel and the development protocol described in this report. They do not represent a final untouched-test evaluation or a claim of effectiveness across all SME sectors. This report is current to 8 November 2026; the project remains in progress until February 2027.

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# *Abstract*

This thesis develops B-SMART, an AI-powered business analytics and decision-support framework intended to connect operational records with performance evaluation, demand prediction, customer insight, and feasible business recommendations for Bangladeshi small and medium enterprises (SMEs). The proposed framework uses a shared operational core with sector- and business-specific constraints. This progress report records the completed research foundation and a first reproducible demand-forecasting experiment using the supplied daily shop–category dataset. The dataset manifest identifies the package as real SME sales or real people; a separately named simulated-unmet-demand table is excluded from the observed-sales target. The analysed panel contains 561,152 daily observations across 32 shops and 16 product categories, forming 512 complete shop–category series from 1 January 2023 to 31 December 2025. Observed sales total 1,580,361 units, and 37.60% of daily observations are zero-sales records. Direct cumulative forecasts were evaluated at 1-, 7-, 14-, and 28-day horizons over three development origins in July, August, and September 2025. A pooled LightGBM Tweedie model achieved WAPE values of 79.88%, 33.87%, 24.22%, and 18.57%, respectively, outperforming the evaluated seasonal-naive, Croston–SBA, and TSB comparators on pooled totals at each horizon. Nevertheless, one-day forecasting remains weak and under-forecasted, while the intermittent-demand subgroup contains only four series and is too small to support a reliable specialist-model conclusion. The current experiment uses historical sales and calendar features only; SKU-level demand, stock-out censoring, price and promotion effects, calibrated prediction intervals, and inventory-policy impact remain subjects for further implementation and evaluation. The results are therefore development evidence, not a final untouched-test claim.

# *Acknowledgements*

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# Chapter 1

# Introduction

## 1.1 Context and Motivation

Operational data is used for business intelligence and analytics to structure the data for decision-making. Decision-makers can explain performance, predict future performance, and select an appropriate response [4]. In a small or medium enterprise (SME), the journey towards analytics starts with familiar records: invoice lines, product movements, purchase receipts, expenses, payments, and customer visits. These records can shed light on revenue and stock position, but their value becomes limited if they remain on paper ledgers, spreadsheets, and separate applications. The owner might be aware of a sales drop last month but not realize what products, customers, or operating conditions were responsible for the reduction, whether the pattern is likely to persist, or whether an inexpensive course of action should be followed.

This is a more challenging decision than making a dashboard. In retail, to ensure product availability, an owner of a business should make an early estimate of product demand and order in advance from a supplier. The order should be determined with respect to limited cash, storage capacity, pack multiples, and delivery time. Too little stock will result in stock-outs and lost sales, while too much will lead to high inventory levels and obsolescence. Much of the capital tied up in excess inventory can lead to near-expiry or expired stock, which can vary based on batch, remaining shelf life, and temperature-sensitive storage. Otherwise, an attractive order may still be possible.

Meanwhile, a once-valuable customer may become inactive, or a product may appear profitable because the costs were incompletely recorded. A promotion might lead to an increase in volume without a corresponding increase in contribution. These questions are based on the same operational history but involve different analytical activities within the same decision context.

Hierarchy, intermittency, promotions, and local events are all identified as factors in retail forecasting research. New products, changing assortment, and organisational use of forecasts are also central practical concerns in retail forecasting [8]. Evidence from large

forecasting competitions also cautions against assuming that a more complex algorithm is automatically more accurate: combinations, statistical methods, and carefully constructed baselines are still competitive in many series [17].

So, the problem investigated in this research is not to put a pretty face on a sales file. It is to construct a defensible path from business events to out-of-time predictions and then from predictions to actionable, explainable decisions.

## 1.2 Bangladesh SME Business Context

The framework targets small and medium enterprises (SMEs) in Bangladesh across business sectors, including retail, wholesale, distribution and services. A shared operational core covers sales, purchases, expenses, payments, customers and performance indicators. Inventory, batch, expiry and capacity modules apply only where relevant to the business. Sector and business characteristics are retained explicitly; a common framework does not imply a universal model or already demonstrated effectiveness in every sector.

Bangladesh-specific studies show that perceived benefit, management and organisational readiness, government support, cost, complexity, and organisational and environmental context influence the adoption of ICT in SMEs [11, 22]. Research on mobile financial services also demonstrates the importance of ease of use and the affordances of everyday practice, such as mobility and self-control [9]. These results suggest that local payment channels and low-friction data entry, together with owner-readable explanations, are requirements for the design rather than optional details for presentation.

This study is based on a functional definition of its population instead of relying on a regulatory size threshold that is subject to change. The set of target businesses has stock, records sales and purchases, makes branch-, product-, or batch-level decisions, and has limited specialist analytics capability. The unit of analysis can be a business, branch, stock-keeping unit (SKU), batch, customer, or historical customer snapshot, depending on the research question. A customer snapshot is either used or not used depending on the specific research question.

Conclusions will be limited to the sampled businesses, sectors and evidence actually collected. Recommendations remain subject to owner approval and business-specific feasibility checks.

## 1.3 Decision Environment and Stakeholder Requirements

An SME analytics artifact is used inside an operating system of people, records, cash pledges, and supplier relationships. The owner or manager is the primary decision-maker, while the evidence could be produced by sales staff, inventory staff, accountants, and external payment or supply services. A recommendation is reliable only if those roles

generate compatible records and the decision-maker can interpret the outcome at the time of action.

The Bangladesh adoption evidence makes the methodological implication clear: perceived value, organisational readiness, cost, and workflow compatibility must be assessed in addition to model accuracy [11, 22].

The setting is translated into concrete information requirements in Table 1.1. The table is a design analysis of the present thesis. It is not a statement that every participating business has the same staffing structure. One individual can fill more than one role in a micro-enterprise.

There is an important difference between those who create an event, those who validate it, and the person who validates the output and suffers the consequences of a bad decision.

TABLE 1.1: Stakeholder decisions, responsibilities, evidence requirements, and corresponding design implications in the target setting.

| **Stakeholder** | **Decisions or Responsibilities** | **Evidence Required** | **Design Implication** |
|---|---|---|---|
| Owner or manager | Reorder products, allocate cash, approve customer contact, and accept or reject a recommendation | Current stock, expected demand, cost range, uncertainty, and supporting reasons | Present a concise and feasible action with drill-down evidence while retaining the owner's decision-making authority |
| Operational users | Maintain product records, minimize duplicate entry, validate imported data, and preserve corrections | Stable product keys, import-validation results, duplicate checks, correction history, and source records | Prevent uncontrolled changes to product keys, reduce duplicate entry, validate records at import, and preserve corrections for traceability |
| Accounts or finance personnel | Reconcile payments, expenses, receivables, and inputs used for margin calculations | Links between invoices and payments, expense periods, cost provenance, and identification of missing cost information | Preserve auditable links among transactions, payments, expenses, and cost records, and explicitly identify incomplete cost evidence |
| Supplier or logistics contacts | Confirm lead time, pack size, minimum-order quantity, incoming quantity, and related supply conditions | Dated purchase orders, receipt history, supplier lead-time records, pack-size information, minimum-order requirements, and incoming quantities | Apply lead-time and ordering restrictions as dated operational constraints rather than as universal constants |
| Research and system team | Build, evaluate, monitor, and maintain the artifact | Versioned datasets, analytical cutoffs, code, parameters, metrics, immutable analytical snapshots, and audit events | Ensure that reported outputs are reproducible from versioned and immutable analytical evidence |

This analysis leads to four owner-facing qualities: that the output is a decision which the business can actually make. Traceability is the ability to trace the value displayed back to its source events, transformations, and a declared cutoff. Proportionality refers to the fact that the model, interface, and data requirements are reasonable for the value and risk of

a decision. The value and risk of a decision is contestable if the owner can examine the supporting evidence, reject the suggestion, and record a reason. These qualities transform local usability from a visual design preference into a condition of credible evaluation.

## 1.4 Problem Statement

The middle problem is the lack of a reliable and convenient link between data from these logs and the analysis of the resulting information. These logs are used for three activities: recording business operations, extracting predictive insights from the data, and choosing an action that is viable for the business.

Orders can be stored in a transaction system, but even when the data are available, there may be no forecasts. A stand-alone model can report an accuracy score after evaluation, but it can still be untrustworthy because the features may not have existed at the time the prediction was supposed to be made. Its test data may have contained information from the training period, or it may be unable to deliver an output that is related to budget, stock, lead time, and customer consent. A recommendation can therefore be technically correct but not feasible for the business.

The problem is broken down into four related gaps:

G1. **Data foundation gap.** Sales, inventory, purchases, expenses, customers, and payments need a consistent schema, timestamps, provenance, and organisational boundaries before they can support reliable and comparable analytics.

G2. **Prediction-validity gap.** Demand and customer-risk models need historical snapshots, chronological or entity-safe splits, training-only transformations, consistent baselines, and metrics suitable for sparse or imbalanced outcomes. Without leakage control, there can be an attractive but false appearance of performance and validity [15].

G3. **Prediction-to-action gap.** A forecast or risk probability does not identify what to order, who to reach out to, or whether an action is cost-effective. Prediction must therefore be related to costs, constraints, and uncertainty. This represents the shift from predictive to prescriptive analytics [1].

G4. **Adoption and accountability gap.** An SME owner requires a short, clear explanation of the reason, confidence indication, and traceable source for a suggestion. A feature-attribution method can explain how features contributed to a model output, but it does not establish causality, nor does it attribute credit or blame for a model's behaviour. Instead of relying solely on human judgement, methods such as SHAP and explanation frameworks provide structured mechanisms for interpreting model behaviour [16, 18].

This thesis therefore investigates the possibility of an integrated framework that is Bangladesh-aware and can visually represent an auditable retail record, a leakage-controlled forecast, customer insights, and ranked actions based on the operating constraints of an SME. The work does not assume that machine learning can outperform every existing baseline. Finding that a simple rule is more reliable or economical for a particular task is considered a valid result.

## 1.5 Analytical Logic and Claim Boundaries

The thesis, which is composed of four logical objects, is often lumped together in a single analytics process. A prototype of an analytics system can be viewed as a sequence consisting of an observed business event, an analytical estimate, an action influenced by that estimate, and a later outcome. Figure 1.1 shows this dependency. There are failure modes associated with each of these transitions. Incomplete or inconsistent records reduce the available evidence; leakage or invalid records invalidate an estimate; ignored constraints make an action infeasible; and without outcome capture, it is not possible to determine whether the recommendation contributed to a successful outcome or to learning. Success in one area must therefore be matched by success in another.

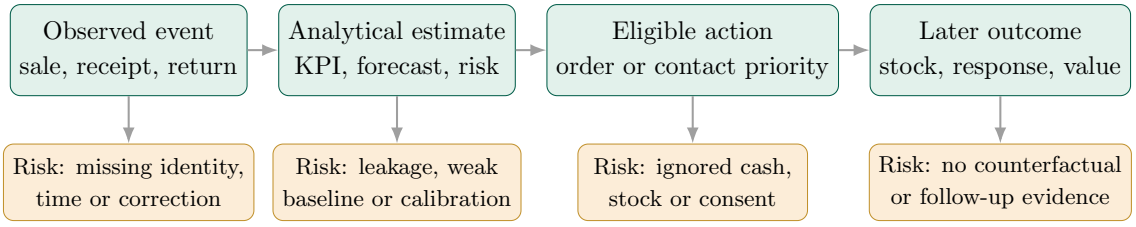



FIGURE 1.1: Logical dependency from an operational event to an evaluated outcome.

This separation also regulates the claims made in the thesis. A dashboard value is *descriptive*: it is supported when the accounting definition and source reconciliation are correct. The likelihood of inactivity indicated by a forecast or inactivity score is *predictive*: it is only available on observations that were not used during model development. A rank recommendation is *prescriptive*: it also calls for specific actions, considers costs, and is planned under constraints and uncertainty [1]. A positive field-impact claim is even stronger, since it relies on what decisions were made based on what was observed and on outcomes within a defensible comparison.

Several terms are therefore used in a narrow sense. *Demand* refers to the quantity required by customers; recorded sales may be a censored observation of demand when stock is not available. *Future inactivity* refers to no observed demand in a future period and is not assumed to be permanent or, where applicable, a "forever" purchase pattern.

A faithful summary of model behaviour should be supported by the behaviour itself and by the decisions made, rather than by assuming that attributed behaviour will necessarily cause the desired outcome [16, 18]. Finally, *local validity* refers to the validity demonstrated

for the businesses involved, the sampled period, and the observed workflow, rather than automatic generalisation to every SME in Bangladesh.

Negative and conditional findings are therefore informative within these boundaries. A model may be accurate but not actionable because of a lack of cost data. An action policy may lead to reductions in simulated stock-outs only under an assumed lead time and may therefore not be deployment-ready. An explanation can be technically faithful but still not be understood by owners. The evaluation is planned to uncover these distinctions. Instead of combining all these conditions into one label of success, it is advisable to evaluate and report each condition separately.

## 1.6 Research Aim and Objectives

### 1.6.1 General Objective

The overall goal is to create and test an AI-based business solution. The analytics framework is designed as a solution that allows Bangladeshi SME businesses to track their business performance, extract reliable and explainable insights, and prioritise feasible strategies while maintaining owner control.

### 1.6.2 Specific Objectives

The study will:

O1. specify an auditable canonical data model for sales, with a focus on the business and its activities, including products, stock, batches, expiry, purchases, returns, customers, suppliers, payments, and expenses;

O2. build reproducible preprocessing and feature pipelines for performance analysis, SKU demand, customer inactivity, and segmentation;

O3. compare transparent baselines, a pooled multi-business model, and shop-adapted models using chronological and held-out-business validation, with both predictive and decision-oriented measures;

O4. measure uncertainty and provide explanations at both global and case levels, while recognising that association does not imply causation;

O5. formulate a mechanism that retains constraint awareness in reorder, retention, and resource-allocation decisions; and

O6. assess technical correctness, robustness, scalability, usability, and potential business value at distinct levels of evidence, using data collected with consent from the Bangladeshi SME sector.

## 1.7 Initial Solution Concept

The proposed solution is B-SMART: the Bangladesh SME Monitoring, Analytics, Recommendations, and Tracking System. It is a multi-sector business decision-support framework with a shared operational core and configurable sector-specific analytical tasks and constraints. It does not assume that one trained model is suitable for all businesses.

SME business events are brought into a governed operational layer, where validated events are converted into time-frozen analytical views. A clear comparison is then made among transparent baselines, a pooled multi-business model, and shop-adapted candidates, where sufficient evidence is available. The selected model produces forecasts, risk estimates, and segment profiles for an active strategy layer. This strategy layer maps the forecasts, risks, and segment profiles to feasible strategies. Before listing and prioritising actions, the shop owner provides the relevant constraints. Each displayed action is accompanied by the data cutoff, model or baseline source, uncertainty or error range, source documentation, and supporting factors. The owner's response and the subsequent outcome are retained in the system for evaluation and monitoring.

The three analytical outputs match the approved title. *Performance evaluation* tabulates customer behaviour, stock movement, and margin, all of which are associated with revenue from reconciled records. *Insight extraction* estimates future demand, identifies meaningful behavioural groups, and detects customer inactivity. These estimates are then used in *strategy optimisation* to generate candidate actions, such as adjusting order quantities and/or retention priorities, subject to cash, storage, lead time, minimum order quantity, current stock, and customer-contact consent.

These components are formally defined in Chapter 3.

Figure 1.2 provides a compact scope map of this proposed decision-support path. It also makes explicit that governance and owner control are not optional add-ons: they constrain every stage from source records to subsequent learning.

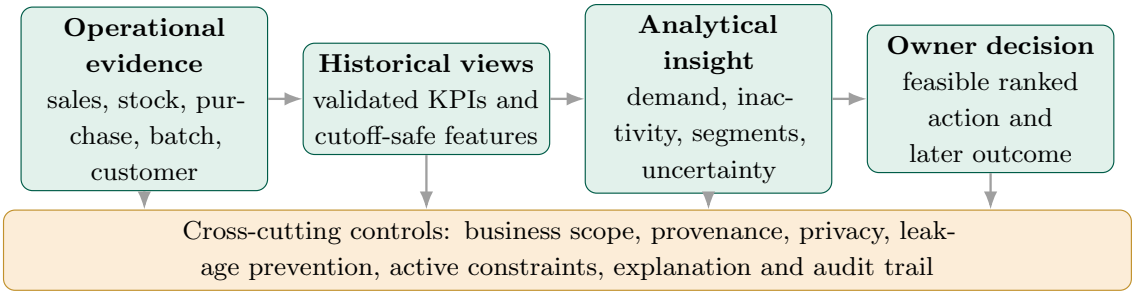



FIGURE 1.2: Functional scope of B-SMART from business evidence to an accountable owner decision.

## 1.8 Research Questions

The thesis is based on four research questions:

**RQ1.**
Do features such as calendar effects, price, promotion, and market context enhance out-of-time branch–SKU demand forecasts over seasonal and statistical baselines in the target business setting?

**RQ2.**
How accurately and reliably can a historical customer snapshot predict whether there will be no repeat purchase within a predetermined future time period?

**RQ3.**
Under the same demand and operating constraints, can a forecast-driven reorder policy reduce stock-outs and holding costs compared with the current policy of the participating business?

**RQ4.**
Do short Bangla action descriptions enhance owner task-friendliness, confidence, and perceived usability without causing information overload or increasing decision time?

## 1.9 Scope and Boundaries

The study focuses on Bangladeshi SME businesses, structured operational data, descriptive indicators, demand forecasting, customer inactivity, behavioural segment identification, explainability, and business strategies such as replenishment, stock-out review, overstock or near-expiry review, and consent-aware customer retention.

Participating businesses contribute only their own portion of the data. Installed and anonymised research fields are used for cross-shop modelling, while the raw records and constraint profile of each shop remain tenant-isolated. External portability testing on a different dataset does not, by itself, validate the portability of the method to Bangladeshi business settings. Public price information can provide background information but cannot replace evidence at the invoice, stock, batch, and constraint levels.

The study does not include manufacturing scheduling, credit scoring, employee surveillance, uncontrolled price changes, or unrestricted customer profiling. It does not claim causal inferences based on attributions of observational features. It will not treat generated data as field evidence or mix synthetic and observed materials without clear identification. Reported real-world scores will not be extrapolated by assuming that they are equivalent to the performance of participating businesses.

The study does not claim to transfer its results automatically to other SME verticals or to non-validated settings. The implementation details, experimental results, and final

conclusions are presented in Chapters 4–6. Any optional feature can be deliberately enabled and evaluated within the defined experimental scope.

## 1.10 Expected Contributions and Significance

The evaluated integration of established research methods, constrained within the context and relevance of small SME businesses, is expected to be a research contribution. Specifically, the work is expected to include (i) a traceable business operational-to-analytical data design; (ii) reusable historical-snapshot protocols designed to avoid common forms of data leakage; (iii) a Bangladesh-aware comparison of forecasting and customer-inactivity approaches; and (iv) a clear link between prediction and constrained, actionable recommendations.

The work also provides evidence of the ability of owners to interpret and apply the information presented to them. When making a recommendation, the system presents supporting evidence to help the owner understand the "how and why" of the recommendation, while using familiar rules and maintaining the owner's control over the action. It is not intended to exclude the owner from the decision or to label a numerical output as profitable without considering the business context. Before measuring the impact on the business, the system must first establish that its outputs are accurate, interpretable, and actionable.

## 1.11 Anticipated Challenges

The most crucial dependency is on relevant primary data obtained from several consenting businesses. The purpose of the aggregated history is to obtain invoice lines, inventory, products, batches, expiration dates, prices, and customers. It may not be easy to estimate the number of products, batches, expiration date data, prices, and customers present in a particular business. Available publicly sourced data in Bangladesh may not contain the identifiers needed, and partner data may be lacking. Change in SKU codes, lack of consistency between box/pack/unit units, and missing cost or measurement data are some reasons for such issues.

The technical risks are intermittent demand, short history, new products, stock-out censoring, class imbalance, and data drift. The scientific risks are optimistic, underwhelming, or unsafe results of the models, incompatible comparison between datasets, conclusion drawn from a single model, and the disruptive to-and-fro process of data between several models. The human risks are time constraint, disparity in digital skills, lack of trust in the digital setting, and excessive confidence in explanations.

The methodology has a prevention or evaluation step associated with each risk.

## 1.12 Organization of the Report

Chapter 2 builds up the technical and contextual background, reviews the relevant literature, and identifies the research gap. Chapter 3 introduces the research design, data protocol, formal tasks, proposed framework, and evaluation plan. The Implementation and Experimental Results chapters are kept as placeholders in this preliminary report. The Conclusions chapter brings together only what can be concluded from the current background study and proposed methodology. Final conclusions based on empirical outcomes will be drawn from the evidence obtained during the implementation and experimental evaluation.

# Chapter 2

# Background Study and Related Work

## 2.1 Background Study

### 2.1.1 Business Intelligence and the Decision Continuum

Business intelligence (BI) is a blend of data, infrastructure, analytical techniques, best practices, experience, and organisational practice used to enhance decision-making. Chen, Chiang, and Storey trace its evolution from enterprise reporting to web, mobile, and sensor-rich analytics [4]. This thesis follows a three-step approach: *descriptive analytics* determines what happened and where; *predictive analytics* predicts what is likely to happen next; and *prescriptive analytics* compares feasible responses.

The stages are interconnected. An action recommendation is only as believable as the prediction that supports it, and a prediction is only as believable as the operational data from which it was obtained. There are also different integrity requirements for operational and analytical workloads. While a sale requires consistent updates to stock and payment records, an analytical model requires a historical feature view containing values that would not change once the evaluation point has been reached.

The proposed framework therefore separates the transaction record from versioned analytical snapshots. It maintains the connection between them through source identifiers, timestamps, and transformation metadata.

### 2.1.2 SME Digitalization in Bangladesh

The decision to adopt technology is not only a matter of the accuracy of technology models. A mixed-method study indicates that the important factors determining ICT adoption among SMEs in rural Bangladesh are awareness of the benefits, government support,

top-management support, and financial support [11]. A later survey of 535 SME managers combines the technology–organization–environment model with diffusion-of-innovation concepts. The authors [22] recognize that factors such as relative advantage, complexity, perceived cost, management support, competitive pressure, and government support are important considerations. These studies suggest that organizational readiness and costs, as well as technological fit and workflow functionality, warrant attention when considering technology adoption. However, the two studies do not evaluate a retail prediction system.

This operating context also includes mobile financial services. Through in-depth interviews, accessibility, spatial and temporal mobility, and access to information were found to be important issues in the daily use of MFS in Bangladesh. Self-control and the affordances related to self-control are also important considerations [9]. Therefore, the proposed data model includes the payment mode in the settlement status field, along with other common business fields. The study should not be construed as providing evidence of better business performance for a named provider or of better performance of one local provider compared with another.

### 2.1.3 Shared SME Core and Sector-Specific Requirements

The research scope covers SME businesses across sectors. Shared entities support comparable performance indicators, while sector-specific extensions capture differences in products, service capacity, sales cycles, procurement and operating constraints. Inventory-based firms may require SKU, stock, pack, batch and shelf-life fields; service firms may require service type, bookings, staff availability and delivery capacity. Fields that do not apply are marked not applicable rather than populated with invented values.

The artifact supports business performance evaluation, insight extraction and feasible strategy recommendations. Owners retain authority over actions. Sector-specific requirements and owner constraints are assessed before an analytical task is enabled; unsupported tasks return an explicit evidence gap.

### 2.1.4 Performance Evaluation and Data Foundations

Well-defined measures are the starting point for performance evaluation. Revenue, gross margin, average order value, stock turnover, return rate, and customer repeat rate each need to include a defined time window, an aggregation method, and appropriate treatment of events such as rescheduling, refunds, and lost fees. If any of these elements are not incorporated into a dashboard value, the resulting measure may be misleading or ignored. Definitions may also differ from the accounting records or may refer to different time periods. The canonical layer therefore maintains invoice lines and stock movements while creating analytical measures through documented views.

Data provenance concerns where a record came from, when it was received, and how it was transformed. Primary business data should be accompanied by provenance information,

together with appropriate customer anonymisation and organisational anonymisation. For public data, the context and reasonable implications of the applicable licence should also be documented.

### 2.1.5 Data Quality, Provenance and Temporal Semantics

There is no single data quality percentage that can adequately describe the quality of a dataset. In this study, data quality is considered across a minimum of five dimensions. *Syntactic validity* asks whether a value has the expected type, range, and format. *Entity integrity* asks whether the keys and valid relationships among invoices, products, customers, and branches are stable. *Temporal integrity* asks whether event, correction, receipt, and availability times are possible and sensible. *Reconciliation* asks whether the movements of payments, returns, and stock are in agreement according to declared business rules. *Decision adequacy* asks whether the fields required for a specific analysis are complete enough at its decision cut-off. A file may score well on one dimension and still be unsuitable for a particular task. Likewise, a dataset may be structurally complete but have no value for a margin analysis if unit cost is unavailable. Similarly, if customer identities are not available for a customer-level study, the relevant analysis cannot be performed. The distinction is significant because the data used to operate the system are continuously updated.

A cancelled invoice can be exported, a return can occur days after the original sale, and a return can later be rejected. Apparent availability history may also be affected by a late stock adjustment. Replacing the original value with a corrected value can therefore eliminate the evidence required to reconstruct what the system knew at an earlier point in time. A more robust design is append-aware: it keeps the source event, records the correction or transition to a new status, and creates a versioned analytical view based on a declared cut-off. This reflects the BI principle that analytical value depends on an understandable connection between data, infrastructure, and organisational use [4].

Missingness also needs to be interpreted rather than automatically imputed. Missing customer identity does not necessarily indicate an unsuccessful import; it may represent an anonymous cash sale. Unit cost may be missing, which makes a record unsuitable for gross-margin analysis. Similarly, missing values may make a record unsuitable for unit-demand forecasting. A zero sale can represent zero demand, a closed location, a missing file, or an unobserved stock-out. These states have different meanings and should be separated when the source provides sufficient evidence to distinguish them. Eligibility is therefore proposed at the level of a research task, business, and period rather than by declaring an entire dataset universally ‘clean” or ‘dirty”.

This quality model is completed by provenance. Each derived field should be identified by its source columns, transformation version, and analytical cut-off. Each reported result should also record the dataset manifest, data split, and model or rule version that generated it. This is not merely corrective inheritance; it is an auditable process that makes the

analytical results reproducible. It also helps ensure that a public or generated dataset is not silently substituted for the primary Bangladeshi business evidence.

### 2.1.6 Customer Analytics: RFM, CLV and Segmentation

A simple summary of *recency*, *frequency*, and *monetary value* is provided by RFM. It summarises a customer's purchase or transaction history using features that are easy to calculate and maintain an unambiguous behavioural interpretation. Fader, Hardie, and Lee link RFM with Customer Lifetime Value (CLV) through the Pareto/NBD and gamma–gamma components, demonstrating how the value of a particular customer history can be estimated [7]. CLV can be used as an outcome or planning measure, but it is not necessarily suitable as a direct prediction target or operational goal.

A field that defines a segment should not be admitted as an independent predictor of that same segment, as this would introduce information leakage. The question for unsupervised segmentation is whether coherent behavioural groups can be identified without an existing label. K-Means aims to:

$$\underset{\{C_k,\mu_k\}_{k=1}^{K}}{\text{minimize}} \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2. \tag{2.1}$$

The procedure returns clusters even when the data form a continuum. Therefore, both the number and the interpretation of the clusters need to be validated. The silhouette coefficient measures the similarity of observations within a cluster and their separation from the closest other cluster [20]. The stability of the clustering parameters across resampling and the business usefulness of the resulting clusters should be treated as separate tests. An attractive label such as "loyal" does not, by itself, imply statistical stability or actionability of a cluster.

### 2.1.7 Retail Demand Forecasting

A demand forecast estimates the number of units needed for a specific product, location, and future horizon. Retail demand series can be influenced by price, promotion, stock-outs, festivals, assortment changes, and aggregation across products or branches. Fildes, Ma, and Kolassa emphasize that retail forecasting practice cannot be simplified into a single algorithm because of data hierarchy, operational requirements, and the amount of data involved. The forecasting process and the way forecasts are used also matter, alongside statistical accuracy [8].

Intermittent SKU demand contains numerous zero-demand periods between non-zero observations. Croston's method smooths the non-zero demand size and the interval between demands separately. Syntetos and Boylan examine the accuracy and bias of intermittent-demand estimates [24]. This work supports the use of an intermittent-demand

baseline rather than requiring a uniform and constant baseline. For dense time series, an appropriate dense-series model can be used at the SKU level. Forecasts also need to be reconciled across different levels of aggregation, including product, category, branch, and overall business totals. Optimal-combination reconciliation provides a principled means of generating coherent hierarchical forecasts [12].

When modelling, the model should be only as complex as the available evidence supports. ARIMA-family models represent serial dependence, including dependence after appropriate transformations; tree ensembles can learn nonlinear effects from lagged, calendar, and price variables; and sequence models can use ordered historical observations. Random Forest and XGBoost provide suitable tabular alternatives [2, 5]. The M5 retail competition illustrates the importance of large-scale, hierarchical, and uncertainty-aware comparison [17]. A seasonal-naive forecast is therefore a necessary benchmark rather than a ceremonial baseline.

Although inventory usefulness is related to forecast accuracy, the two are not the same. Hyndman and Koehler discuss the limitations of common percentage-error measures and suggest alternative approaches, including MASE as a scale-free measure for comparison [13]. A study by Syntetos and colleagues associates the choice of forecasting methods with stock-control requirements, whereas forecasting selection based solely on statistical accuracy may not reflect the needs of inventory management [23]. The methods are therefore evaluated using point-forecast error, uncertainty, and simulated stock-out and holding costs rather than selecting a technique based on a single easily computed metric.

### 2.1.8 Forecastability, Demand Regimes and Decision Loss

The forecasting task varies according to the information content of a series. A high-volume staple that is repeatedly sold each week can provide many observations of its level and seasonality. In contrast, a slow-moving spare item can contain many zero-demand periods, while a newly introduced product may have little or no direct historical data. Pooling these different cases into a single score and evaluating them with one algorithm can mask systematic failures. The literature on intermittent demand therefore suggests an approach based on the demand regime. Comparisons should consider zero-demand frequency, non-zero demand variability, and data availability when determining the eligibility of a forecasting method [6, 24].

This gives rise to another trade-off: temporal aggregation. Daily data preserve the temporal order of observations and can support short-range ordering decisions, but they may not always provide sufficient stability for model estimation. Aggregating data on a weekly basis can reduce sparsity, but it may also mask festival peaks and other short-term patterns that are relevant to stock-out intervals and short lead-time decisions. Product-level aggregation can provide more signal, but an SKU-level forecast cannot be directly determined from a category-level forecast. Hierarchical reconciliation addresses coherence between different aggregation levels without removing the need to choose a granularity that corresponds to the operational decision.

Availability is limited, and observed sales are therefore not always an ideal target. If an item is not in stock, a zero or low recorded sale does not constitute proof that customer demand was zero or low. Training on such periods without a stock-availability indicator can teach a model that low demand is associated with the observed supply constraint. When censored demand periods exist, the analysis should therefore consider how the results change when those periods are excluded or when alternative assumptions are applied. Where sufficient history exists, sensitivity analysis can be conducted under different treatments of censored periods. When there is no way to reconstruct the underlying demand, the study should avoid claiming that the observed sales represent unconstrained demand.

Forecast errors also have asymmetric decision consequences. Under-forecasting can lead to stock-outs and lost sales, whereas over-forecasting can lead to excess inventory and holding costs. The cost of one decision or alternative may therefore differ substantially from that of another. These costs cannot necessarily be minimised by selecting the model with the lowest MAE, particularly when forecast errors affect other products or future periods. This is why scale-aware error measures, prediction intervals, and inventory simulations using identical demand paths complement one another in their respective purposes [13, 23]. Model comparison can remain prediction-focused until the relevant cost assumptions are declared, after which decision-oriented comparisons can be conducted.

The review therefore rejects three shortcuts: first, selecting a deep model before establishing the structure of the data; second, assuming that a simple model is always sufficient; and third, treating model complexity itself as evidence of value. The analysis should instead begin by measuring the structure of the series and comparing appropriate baselines and candidate models. A result from a large-scale forecasting competition should not be treated as direct evidence of business value for a small local retailer. The M5 competition provides strong evidence for comparative retail forecasting, but the present task is to determine which aspects of that evidence transfer to a setting with substantially different data volumes, infrastructure, and decision costs [17].

### 2.1.9 Customer Inactivity, Class Imbalance and Calibration

Churn can be a documented cancellation for a subscription-based service. For non-contractual retail, however, churn has to be specified at a cutoff time $t$ as no repeat purchase within a given time interval $[t, t + H]$. All features must be known at time $t$, and the training and validation processes must not be contaminated by information from the future label window. This definition translates a business term into a repeatable prediction task.

In business-to-business settings, the positive class can be scarce or vary across datasets. Saito and Rehmsmeier provide a rationale for the superiority of precision–recall analysis over ROC analysis for imbalanced binary classification tasks [21]. Model comparison should therefore report class prevalence and the quality of the predictions before considering action costs and benefits.

The prescriptive layer includes an additional requirement. When using a model output in a decision calculation, ranking alone is not sufficient; the output needs to be converted into an interpretable probability or score. A probability of 0.7 should, under appropriate calibration, correspond approximately to a 70% frequency of the event among comparable cases. Niculescu-Mizil and Caruana examine the calibration of probability estimates from supervised learners and post-hoc calibration methods [19]. The proposed framework therefore reports Brier score in addition to PR-AUC, ROC-AUC, and top-fraction lift, while also making calibration information available.

### 2.1.10 From Customer Score to Responsible Retention Decision

A customer-risk model does not grant permission for an action; instead, it provides a ranking or probability. For a given contact capacity, ranking quality determines whether the selected fraction contains more future inactive customers than an untargeted policy. When an expected value is calculated, the probability needs to be calibrated because the benefits and costs are subject to uncertainty and are weighted by the estimated probability. Therefore, the operational threshold should be determined using validation data and the pre-declared assumptions. The final test set should not be used to determine contact capacity or utility assumptions; rather, it should be used to confirm the choices made during the development process.

Additional ambiguity is created by the customer record. The frequency and amount of purchases may be linked to opportunities to visit, household purchasing patterns, or shared identifiers. Neither return behaviour nor discount-seeking behaviour alone is sufficient to indicate loyalty or future value. Fader, Hardie, and Lee demonstrate why combined transaction history can provide more information than a single RFM label [7]. Segmentation can support interpretation, but a persuasive "segment name" is only a description of a behavioural pattern. It does not establish that the pattern is statistically stable. Therefore, clustering should be accompanied by stability checks and owner review before the resulting segment output is used for decision-making [20].

An independent feasibility gate is also required for responsible use. A score indicating a high probability of inactivity does not, by itself, justify customer contact if consent, contact policy, or an appropriate offer is absent. The score should also never be interpreted as a causal attribution, such as "long recency increased risk," or restated as a causal claim that a particular message will retain the customer. The thesis therefore treats the prediction of inactivity, the reliability of the estimated probability, contact eligibility, and subsequent customer response as separate links in the decision process. This structure allows the model to be scientifically informative even when the available evidence is insufficient to support a claim about profit or the causal effect of a retention action.

### 2.1.11 Leakage, Explainability and Change Over Time

Data leakage occurs when training is exposed to information that would not legitimately exist at prediction time. It can enter through target-derived features, customers who appear on both sides of a data split, post-customer-action information, post-outcome variables, or transformations that are fitted before the data are split [15]. Leakage control is therefore a characteristic of the data pipeline, not a model option. Before training, the system should define the prediction time, feature-availability rules, and data-splitting policy.

SHAP provides additive feature contributions relative to a reference value and supports both local and global inspection of model predictions [16]. Such an attribution describes how the fitted model uses encoded features; it does not establish a causal relationship. In social-science research, as synthesized by Miller, the focus is often on useful explanations that are selective and contrastive [18]. Accordingly, a recommendation card should respond to the question, "Why this action now?" by presenting a small number of salient, decision-relevant factors. It should avoid causal language while providing sufficient supporting evidence for the owner to assess the recommendation.

Business conditions also evolve over time. Retail monitoring therefore needs to distinguish between data-quality failure, input-distribution shift, and deterioration in outcome performance. A model should not be considered valid simply because it remains on a retraining schedule. Evidence that a model remains valid should come from continued evaluation under a declared validation protocol. Any retrained model should be evaluated using the same validation principles that were established during development, with changes explicitly documented.

Once a prediction is made, the next step is to take an action subject to business constraints. Prescriptive analytics incorporates predictions into an explicit decision problem. Bertsimas and Kallus develop prescriptions based on data with auxiliary information, where observations are used to inform conditional stochastic optimization [1]. The present thesis follows the underlying principle of evaluating actions according to expected benefit, cost, and constraints. It does not claim to reproduce the asymptotic method or theoretical approach of the cited work. Instead, reorder and retention decisions are treated as separate action families because their outcomes, costs, permissions, and uncertainties differ.

The proposed system is both an information-system artifact and an analytical study. Design-science research develops and evaluates artifacts in relation to a relevant problem and an existing knowledge base [10]. This perspective contributes to an evaluation plan that considers predictive validity, software behaviour, usability, and the consequences of decisions made by the owner, rather than treating a single model metric as the definitive criterion of success.

### 2.1.12 Cross-Stream Evidence Synthesis

There is no single body of literature that provides the full argumentation for the proposed artifact. Bangladesh-focused technology-adoption studies do not address the complete transaction-level analytical workflow. Forecasting and customer analytics typically begin after an analytical dataset has been produced and provide task-specific methods for constructing models. Research on leakage, calibration, and explanation provides important safeguards, but it does not define an SME operating workflow. Prescriptive analytics provides a connection between predictions and estimates and the decisions that follow from them. Design-science research further requires that the resulting artifact be tested and evaluated against the problem it was designed to address. Figure 2.1 demonstrates how these bodies of evidence address different aspects of the problem and therefore should not be treated as interchangeable requirements.

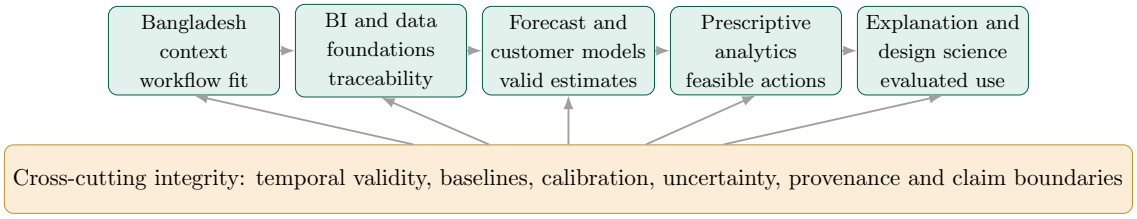



FIGURE 2.1: Complementary roles of the literature streams in the proposed design.

The arrows do not imply that the process is strictly one-way. The local workflow can constrain model choice through data adequacy, which may require changes to the available data fields. Uncertainty can rule out an otherwise attractive action, while owner assessment can reveal the infeasibility of an explanation or recommendation. The figure instead represents logical dependency: later claims require earlier evidence to be credible. The cross-cutting integrity conditions apply to all stages, whereas other conditions may be relevant to only the first, second, or third stage.

As a result of this synthesis, a more rigorous definition of integration is obtained. Integration is not simply the combination of a dashboard, a forecast, and a classifier in a single user interface. It requires structural integration across the system's hardware and software components, as well as a common operational language. Several changes are therefore made to the English representation of the framework, including the addition of explicit cut-offs, task eligibility, separate uncertainty statements, and a single point of auditability throughout the process, from input to outcome. These properties become evaluation targets in Chapter 3.

## 2.2 Literature-Review Procedure

The current literature review is intended to be a structured background review rather than an exhaustive systematic review. Literature sources were chosen based on their

direct relevance to one out of six topics: Bangladesh SME digitalization, BI and design science, customer analytics, retail demand/inventory, churn and evaluation integrity, and explainable or prescriptive analytics. Searches were done using keywords like *Bangladesh SME ICT adoption*, *retail demand forecasting*, *intermittent demand*, *RFM customer segmentation*, *churn profit*, *data leakage*, *probability calibration*, *concept drift* and *prescriptive analytics*.

The sources were preferably taken from peer-reviewed journals or conferences. Literature sources were kept if they defined a concept needed for the topic, provided a method or an evaluation criterion or contained evidence specific to Bangladesh. Metadata of each DOI was verified against Crossref, and dataset sources were verified against DataCite or the repository itself. Only specific numerical statements that could be found either in the metadata, abstract or full text of the source were considered. The core set contains 22 academic publications, and there are 3 candidate datasets mentioned in separate list in Chapter 3. Thus, there are 25 citations in this initial report.

### 2.2.1 Business and Bangladesh Evidence

The BI literature provides the descriptive–predictive–prescriptive framework [4], while design science provides an artifact-oriented research logic [10]. Bangladesh-specific studies further highlight the importance of organizational support, perceived value, cost, complexity, and actual technology use within the local context [9, 11, 22]. These factors should therefore be considered in the design and evaluation of the proposed system.

However, the primary limitation of these studies for the present research lies in their unit of analysis rather than in their methodological quality. Adoption surveys and interview-based studies can explain organizational readiness, perceived benefits, barriers, and technology-use behaviour, but they do not provide transaction-level evidence for evaluating demand forecasting, customer inactivity, inventory decisions, or other operational outcomes. Consequently, these studies are used primarily to establish contextual requirements and evaluation considerations for Bangladeshi SMEs, rather than as direct evidence of predictive or operational effectiveness.

### 2.2.2 Customer Evidence

Customer analytics literature provides several complementary forms of evidence. Probabilistic customer models, such as the framework proposed by Fader et al., establish a relationship between historical transaction behaviour and the probability of future purchasing activity. Such models provide a theoretical basis for analysing customer recency, frequency, value, and inactivity patterns.

For behavioural segmentation, the silhouette coefficient provides an internal measure of cluster cohesion and separation [20]. It can therefore be used as an additional diagnostic for evaluating whether candidate customer segments are internally consistent and sufficiently

distinct from one another. However, a strong silhouette score alone does not demonstrate that a cluster is commercially useful, stable over time, or causally meaningful.

Together, these approaches motivate behaviour-based customer profiling and inactivity analysis. Nevertheless, they do not by themselves constitute a leakage-safe, temporally validated, and operationally actionable framework for non-contractual Bangladeshi retail customers. The proposed system therefore extends these principles by evaluating customer evidence at explicit historical cutoffs and by separating predictive performance from subsequent business action and outcome evaluation.

### 2.2.3 Forecasting and Inventory Evidence

Forecasting and inventory research provides the methodological foundation for evaluating demand predictions and their operational consequences. The M5 competition offers large-scale comparative evidence on retail forecasting methods and evaluation practices [17], while the retail forecasting literature provides additional guidance on the characteristics and evaluation of real-world retail demand [8].

Intermittent-demand methods are particularly important when sales occur sporadically or contain many zero-demand periods [6, 24]. Such settings require comparators that explicitly account for the timing and magnitude of demand rather than relying only on models designed for dense and regular time series. Similarly, hierarchical forecasting and reconciliation methods emphasize the need for consistency across related aggregation levels, such as SKU, category, and branch forecasts.

Stock-control research further demonstrates that forecasting accuracy should not be treated as equivalent to inventory performance [23]. A forecasting model may achieve a lower statistical error without necessarily reducing shortages, holding costs, or cash tied up in inventory. Therefore, an SME-oriented evaluation should go beyond a single forecasting model or a single error metric. Forecast candidates should first be compared using appropriate out-of-time predictive measures and relevant baselines, after which their outputs should be evaluated through inventory or replenishment decisions under explicitly stated operational assumptions and constraints.

### 2.2.4 Risk, Integrity, and Action Evidence

Risk-oriented predictive tasks require evaluation beyond conventional ranking accuracy. Precision–recall analysis is particularly relevant when the target class is rare or imbalanced, while probability calibration evaluates whether predicted probabilities are numerically reliable [19, 21]. These two perspectives address different properties of a risk model: discrimination determines whether higher-risk cases are ranked appropriately, whereas calibration determines whether the associated probability estimates can be interpreted reliably.

Data leakage introduces a separate threat to model validity. Leakage research shows that apparently strong predictive results may become unusable when features, labels, transformations, or data-processing procedures incorporate information that would not have been available at the actual decision time [15]. Consequently, temporal availability and provenance must be treated as explicit conditions of model evaluation rather than as post-processing checks.

Model interpretation and human-centred explanation provide another layer of evidence. SHAP offers a unified framework for local feature attribution [16], while human-centred explanation research emphasizes that useful explanations should support inspection, comparison, and communication rather than merely display model internals [18]. Such explanations can help users understand the factors associated with a prediction, but they should not be interpreted as evidence of causality.

Finally, prescriptive analytics provides the conceptual bridge from prediction to action by incorporating model outputs into explicit cost, resource, and constraint structures [1]. The remaining research question is therefore not whether these individual principles exist, but whether forecasting, customer-risk estimation, leakage control, explanation, and constrained decision support can be integrated into a coherent, low-resource SME workflow and evaluated using evidence aligned to the same temporal boundaries.

## 2.3 Research Gap and Thesis Positioning

The research gap is derived from the difference between what each stream of the reviewed literature establishes and what is required by the target SME decision problem. This gap is deliberately bounded by the 22-work core scholarly set reviewed in this thesis. Therefore, the thesis does not claim that no related system exists anywhere; rather, it identifies six adjoining gaps that remain unresolved within the reviewed evidence base and are directly relevant to the proposed research context.

**G1—Contextual and Empirical Gap.** Bangladesh-focused research explains important determinants and lived affordances of digital technology adoption, including organizational readiness, perceived value, cost, management support, and everyday technology use. However, such adoption-oriented studies do not jointly and longitudinally evaluate SKU-level demand, customer inactivity, and constrained retail actions using transaction-level operational data. Conversely, large-scale retail and forecasting studies provide strong analytical evidence [8, 17], but their validity does not automatically transfer across differences in scale, market structure, infrastructure, and owner-managed operating conditions characteristic of Bangladeshi SMEs. Consequently, evidence that combines local contextual validity with transaction-level analytical evaluation remains limited.

**G2—Operational-to-Analytical Integration Gap.** BI and CRM research demonstrates how operational data can support multiple analytical functions [4]. However, many

forecasting, segmentation, and churn studies represented in the literature matrix begin with a pre-prepared analytical dataset. They do not necessarily examine the complete path from an auditable business event, through a historically valid snapshot, to prediction, action, and subsequent outcome.

This missing path is particularly important for the target business context. Corrections to invoices, stock adjustments, changes in organizational records, and other operational updates can alter the data from which analytical features are derived. If these changes are not versioned and historically traceable, previously generated rows, features, and analytical cutoffs may become difficult or impossible to reproduce. Therefore, an integrated mechanism is required to preserve lineage from source events to analytical outputs and subsequent decisions.

**G3—Methodological-Validity Gap.** Previous research has established several methodological risks and safeguards separately. Data leakage has been examined in [15]; intermittent-demand forecasting has been addressed in [6, 24]; hierarchical forecast coherence has been studied in [12]; forecast-error measurement has been discussed in [13]; and probability calibration has been examined in [19].

What remains missing for the target SME setting is a unified evaluation protocol that applies these safeguards together. Such a protocol should enforce decision-time feature availability, temporal- and entity-safe data splitting, baseline eligibility according to series characteristics, coherent aggregation where required, probability calibration, and a pristine final test set. Without these safeguards being applied jointly, apparently strong analytical performance may not constitute valid evidence of real-world predictive effectiveness.

**G4—Prediction-to-Decision Gap.** Accurate forecasts, customer-risk scores, and customer rankings are intermediate analytical outputs rather than decisions themselves. Stock-control research demonstrates that operational consequences can change the preferred forecasting model even when predictive accuracy appears similar [23]. Likewise, prescriptive analytics formalizes the use of predictive outputs within explicit decision problems [1].

The unresolved requirement is therefore a clear, low-resource mechanism that converts the outputs of multiple SME analytical tasks into feasible actions. Such actions must account for operational constraints including available cash, stock position, lead time, reorder rules, customer-contact constraints, consent, predictive uncertainty, and the cost of taking an action. The gap is not merely to generate predictions, but to translate them into actions that remain feasible within the operating conditions of the SME.

**G5—Owner-Centred Explanation Gap.** SHAP provides principled feature attributions for complex predictive models, while human-centred explanation research shows that useful explanations should be selective, contextual, and aligned with the decision being made [18]. However, neither technical feature attribution alone nor the presence of a generic dashboard demonstrates that a Bangladeshi SME owner understands the

underlying evidence, can distinguish association from causation, or can correctly use the explanation to support the intended decision.

Explanation quality should therefore be evaluated through owner-centred tasks rather than inferred from the presence of an explanation chart. Relevant evaluation should consider whether the explanation presents appropriate supporting evidence, source information, uncertainty, timing, and confidence in a form that enables the owner to interpret and act on the recommendation appropriately.

**G6—Lifecycle-Accountability Gap.** Relationships between business inputs, model outputs, actions, and subsequent outcomes can change over time. Design-science research emphasizes that an artifact must be evaluated in relation to the problem it is intended to address [10]. However, the reviewed task-specific studies do not collectively close the local operational loop by maintaining the relationship among model versions, baselines, recommendations, owner decisions, subsequent outcomes, and revalidation decisions.

Without such lifecycle accountability, a system cannot reliably determine whether a previously generated recommendation remains appropriate after business conditions, input distributions, operating constraints, or model behaviour have changed. A complete SME analytical system therefore requires explicit versioning, monitoring, outcome recording, and revalidation mechanisms.

> From the evidence reviewed, the central research gap is the absence, within the bounded scholarly evidence base, of a jointly evaluated and Bangladesh-centred multi-sector SME system that connects auditable operational records from multiple shops with leakage-controlled and calibrated analytical models; compares appropriate baselines, pooled business models, and eligible business-adapted models; and translates those analytical outputs into shop-constrained, owner-understandable actions whose use and subsequent outcomes can be monitored over time.

Figure 2.2 groups the six gaps into three connected bridges. This grouping clarifies why a model-only contribution would be insufficient: local evidence, methodological validity and accountable use must all be evaluated before the artifact can support a bounded effectiveness claim.

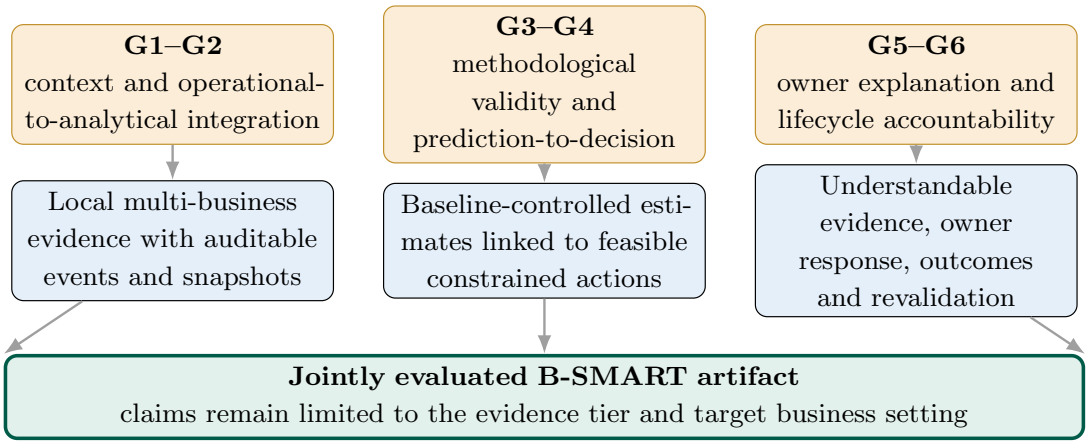



FIGURE 2.2: Research-gap bridge from six literature limitations to the integrated thesis response.

The central gap is therefore integrative and evaluative rather than dependent on the creation of an entirely new classifier or forecasting algorithm. The research must demonstrate whether existing methods can be integrated into a coherent SME decision-support workflow without violating data lineage, temporal validity, decision feasibility, or human accountability.

Table 2.1 makes each identified gap operationally testable by linking it to the proposed system response, the evidence required for closure, and the corresponding study objectives or research questions.

TABLE 2.1: Identified research gaps, proposed responses, and evaluation evidence.

| **Gap** | **Proposed Response** | **Evidence Required to Close the Gap** | **Study Link** |
|---|---|---|---|
| G1: Context | Explicit external-data boundaries and a Tier-1 multi-business study | Primary business records, explicit separation of local evidence from foreign or generated evidence, and clearly bounded cross-context claims | RQ1–RQ4 |
| G2: Integration | Canonical operational events, data manifests, and versioned historical snapshots | Reproducible traceability from source records to features, analytical outputs, actions, and subsequent outcomes | O1–O2 |
| G3: Validity | Temporal- and entity-safe splits, a baseline ladder, and a consistent frozen final test | Out-of-time performance metrics, ablation analysis, uncertainty intervals, calibration assessment, and leakage audit | RQ1–RQ2 |
| G4: Decision | Constraint-aware reorder and retention actions with same-path policy comparison | Operational outcomes under the same demand or customer history, together with sensitivity analysis under alternative cost and constraint scenarios | RQ3 |
| G5: Explanation | High-level Bangla explanations containing supporting evidence, source information, uncertainty, timing, and confidence | Owner-task performance demonstrating understanding and appropriate use of explanations in decision making | RQ4 |
| G6: Lifecycle | Versioning, drift monitoring, response monitoring, outcome tracking, and a formal revalidation protocol | Revalidation logs, model and baseline histories, post-decision observations, and records of recommendation outcomes | O6 |

B-SMART is designed to address this combined contextual, analytical, and operational gap. Public or generated evidence may be used to test selected components of the methodology, but such evidence cannot by itself close the contextual, decision-oriented, or owner-centred gaps. Closing those gaps requires appropriately consented business data and participant-based evaluation from the target context.

Accordingly, the final thesis should report the extent to which such local evidence is available and should distinguish clearly between gaps that are empirically closed, gaps that are only partially addressed, and gaps that remain open. The existence of a prototype, successful benchmark result, or external dataset should not be treated as evidence that a corresponding local research gap has been closed. Benchmark evidence may establish

methodological or technical performance, whereas claims of local effectiveness require evidence from the intended operational context.

# Chapter 3

# Proposed Methodology

## 3.1 Multi-Sector Applicability and Validation

The intended scope is SME business analytics across sectors, rather than one industry. A common core models organizations, branches, transactions, expenses, payments, customers and suppliers. Configurable sector extensions preserve differences in products, services, inventory, resource capacity and operating rules. Demand and reorder experiments apply to inventory-based firms; service firms require service-demand and capacity tasks. Customer-risk tasks require stable pseudonymous identifiers and adequate follow-up in either setting.

Sampling and reporting will retain sector and business identifiers. Results will be reported per business and per sector before aggregation. Pooled models will be compared with sector-specific and business-adapted alternatives using the same chronological cutoffs. Held-out-business tests assess transfer within represented sectors; held-out-sector tests assess transfer across sectors when enough independent sectors and businesses are available. In their absence, cross-sector transfer remains untested. Preprocessing is fitted only on training data, and firms with many rows must not dominate business-level conclusions. No universal accuracy or business-value claim follows from framework coverage.

## 3.2 Research Approach

### 3.2.1 Research Strategy and Design-Science Rationale

This thesis adopts a design-science research approach supported by empirical evaluation. Design science is appropriate because the study develops an artifact intended to address an applied organizational problem and evaluates that artifact against explicitly defined requirements [10]. The principal artifact is the B-SMART framework, including its canonical operational data design, time-aware analytical pipeline, constraint-aware recommendation process, and mechanisms for presenting supporting evidence to the owner.

Empirical evaluation is necessary because a plausible technical architecture alone is insufficient to establish predictive validity, business usefulness, operational feasibility, or usability. Accordingly, the research evaluates not only whether the proposed system can be implemented, but also whether its analytical outputs remain temporally valid, whether its recommendations respect the operating constraints of participating businesses, and whether intended users can interpret and appropriately use those recommendations.

### 3.2.2 Research Stages and Units of Analysis

The research is organized into four related stages:

P1. **Problem and evidence definition:** define the target business-SME setting, research questions, literature-derived requirements, admissible forms of evidence, analytical units, and success measures.

P2. **Artifact design:** define the operational schema, historical snapshots, candidate analytical techniques, policy and strategy constraints, interfaces, and audit records required by the proposed system.

P3. **Technical evaluation:** evaluate data integrity, predictive performance, calibration, robustness, reproducibility, and scalability using explicitly separated development, public-real, and primary-real evidence.

P4. **Decision and user evaluation:** evaluate simulated or pilot operational consequences of inventory and customer-related recommendations and assess whether participating owners can correctly understand and appropriately use those recommendations.

The phase sequence and its evidence gates are summarized in Figure 3.1. Later phases may reveal a design defect and therefore feed back to artifact design, but no feedback cycle is allowed to alter the already frozen final evaluation result.

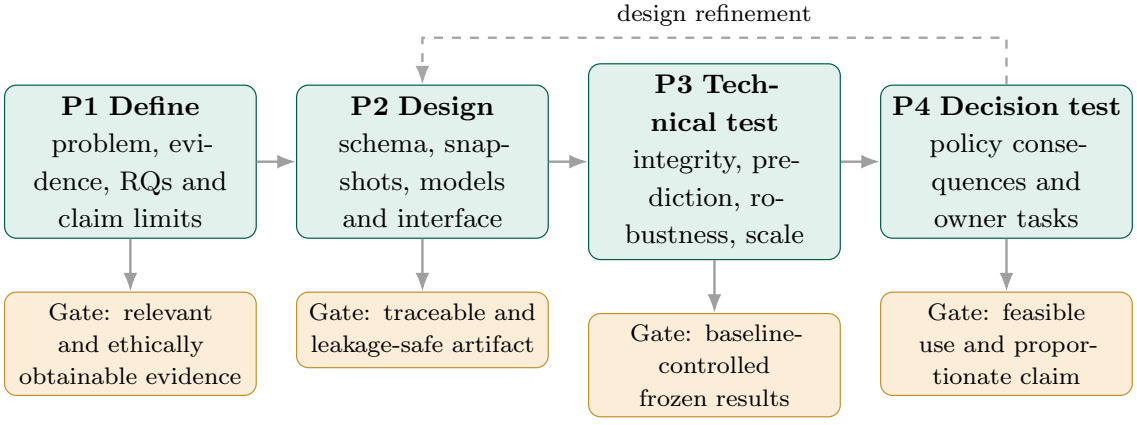



FIGURE 3.1: Design-science research lifecycle and the evidence gate attached to each phase.

The unit of analysis differs according to the analytical task. For demand forecasting, the primary unit is a business–branch–SKU–day or business–branch–SKU–week observation, selected according to the density and reliability of the available data. Batch and expiry information is retained for inventory-related decisions even when demand itself is forecast at the SKU level.

For customer-risk modelling, the analytical unit is a customer feature snapshot constructed at a declared historical cutoff. For strategy evaluation, the unit is an eligible action evaluated under the same information set and relevant product-, business-, or customer-specific constraints as the comparison policy. These task-specific units prevent evidence from one analytical level from being incorrectly interpreted as evidence at another level.

### 3.2.3 Design Requirements and Acceptance Logic

Eight design requirements are derived from the literature review and the problem formulation. They are specified before implementation so that the artifact can be evaluated against stable objectives rather than against whichever functions are easiest to demonstrate.

**DR1: Decision relevance**
Each analytical output shall identify its intended user, decision, analytical unit, decision horizon, and feasible response. An output for which no decision has been specified remains exploratory and shall not be presented as an actionable recommendation.

**DR2: Operational integrity**
Source events shall preserve business and branch scope, stable product, batch, and invoice identifiers, event time, receipt or ingestion time, expiry information where available, transaction status, and correction history. KPIs, analytical features, and model outputs shall be reproducible from these governed source events.

**DR3: Temporal validity**
Features, preprocessing parameters, labels, and model inputs shall use only evidence that would have been available by the relevant historical cutoff. Final evaluation shall be performed on a future period or entity group that has not been used for model or hyperparameter selection [15].

**DR4: Proportionate modelling**
Every analytical task shall begin with an explicit baseline. A more complex candidate model shall be preferred only when the available evidence supports its complexity and when the resulting improvement is material, stable, and relevant to the intended decision.

**DR5: Uncertainty and reliability**
Forecast intervals, probability calibration, data-quality status, and sensitivity to uncertain costs or assumptions shall be reported where those factors could alter the resulting decision [19].

**DR6: Feasible and responsible action**

Recommendation generation shall account for purchasing budget, current and incoming stock, lead time, pack size or minimum-order quantity, storage capacity, remaining shelf life, expiry risk, temperature-controlled storage requirements, contact consent, and relevant policy constraints. Infeasible actions shall be filtered or explicitly withheld rather than merely assigned a low score.

**DR7: Human authority and understanding**

Recommendations shall provide a concise reason, supporting evidence, source, uncertainty where relevant, and an appropriate alternative. The owner shall remain able to inspect, reject, modify, or postpone the proposed action. Explanation quality shall be evaluated through decision tasks rather than inferred from the mere presence of an attribution or explanation chart [16, 18].

**DR8: Accountability and reproducibility**

Every released analytical output shall retain the versions of the dataset, historical snapshot, feature pipeline, model or rule, operational constraints, and interface associated with that output, together with the owner's response and any subsequently observable outcome.

Acceptance is intentionally non-compensatory. High predictive accuracy cannot compensate for data leakage, privacy failure, or an operationally infeasible action. Similarly, a usable interface cannot compensate for an invalid or unreconciled KPI. Conversely, a candidate does not need to be the most complex model available in order to satisfy the artifact requirements. A simpler baseline may be accepted when it is temporally valid, sufficiently accurate for the decision, more stable, and easier to maintain.

Each requirement will therefore be classified as *met*, *partially met*, *not met*, or *not evaluable with the available evidence*. The final category is important because the absence of appropriate cost, consent, constraint, or follow-up data shall reduce the strength of the conclusion rather than being converted into an assumed success. Task-level metrics will be reported together with requirement-level findings so that a single aggregate score cannot conceal a critical failure.

### 3.2.4 Research Questions, Hypotheses, and Measures

The research questions introduced in Chapter 1 are operationalized as follows:

**RQ1 / H1**

A business demand model incorporating locally available calendar, price, promotion, and market-context features will reduce out-of-time WAPE or MASE relative to both an otherwise identical model without those feature groups and a seasonal-naive benchmark.

**RQ2 / H2**

A model trained from historically valid customer snapshots will predict future inactivity better than an event base-rate or rule-based baseline, as measured by PR-AUC, and will achieve Lift@10% greater than 1.0 without unacceptable deterioration in probability calibration.

**RQ3 / H3**

Under identical demand paths, costs, and operational constraints, the proposed forecast-based replenishment policy will reduce the combined consequences of stock-outs and inventory holding relative to the current replenishment policy used by the participating owner.

**RQ4 / H4**

In counterbalanced owner decision tasks, an explanation-supported condition will improve decision accuracy or confidence without causing a practically detrimental increase in task-completion time relative to an output-only condition.

TABLE 3.1: Alignment of research questions, evidence, comparisons, and primary measures.

| RQ | Unit and Comparison | Primary Measures | Evidence Required |
|---|---|---|---|
| RQ1 | Unseen branch–SKU periods; context-aware model versus ablated models and forecasting baselines | WAPE, MASE, interval coverage, and related out-of-time forecast measures | Time-stamped sales, stock, price, promotion, calendar, and relevant context data |
| RQ2 | Future customer outcome windows; predictive model versus base-rate or rule-based baseline | PR-AUC, Lift@10%, Brier score, and calibration measures | Pseudonymous customer transactions with historically valid feature cutoffs and sufficient future follow-up |
| RQ3 | Identical demand paths; proposed replenishment policy versus the participating owner's existing policy | Stock-out units, inventory holding, and total cost under declared assumptions | Stock records, lead times, purchasing constraints, and validated cost assumptions |
| RQ4 | Equivalent decision situations; explanation-supported output versus output-only condition | Decision correctness, completion time, confidence, and perceived usability | Consenting target users and counterbalanced decision tasks |

The final test set will not be inspected during hypothesis development, model selection, feature selection, or primary metric selection. Measures introduced after this stage will be identified as exploratory. Furthermore, a statistically detectable difference will not automatically be interpreted as practically important unless its magnitude, operational consequence, or cost implication is material to the target decision.

### 3.2.5 Target Population and Sampling Plan

The target population consists of SMEs operating in Bangladesh across retail, wholesale, distribution and service sectors. Recruitment will seek multiple independent businesses and record sector, size and operating model. An initial three-to-five-business pilot can test collection and usability, but cannot establish population-wide or cross-sector effectiveness. Broader sector comparisons require adequate independent businesses in each stratum.

The intended analytical target is approximately 12–24 months of longitudinal records across participating businesses, subject to data availability, data quality, owner consent, and ethics approval. A business with substantially less history may still contribute to testing data collection, mapping, schema, and data-quality procedures, but such a business shall not independently determine claims of temporal or cross-business generalization.

Businesses will be purposively recruited because participation depends on access to relevant longitudinal evidence, including invoice, stock, purchase, batch, and expiry information where available, together with owner agreement. At a minimum, included transaction data shall contain interpretable dates, product identifiers, quantities, and sufficient contextual information to reconstruct the relevant business process. Participating businesses shall also have an owner or responsible operator capable of describing the current replenishment process.

Customer-level analysis imposes additional eligibility requirements. It requires stable pseudonymous customer identifiers and sufficient follow-up after each historical cutoff to determine the relevant outcome. If a business, analytical task, or time period does not satisfy the minimum evidence requirements for a specific analysis, that unit will be excluded from that analysis rather than being repaired through invented identifiers, fabricated constraints, or assumed outcomes.

### 3.2.6 Data Sources and Evidence Tiers

Evidence will be separated into explicit tiers so that convenient external, public, or generated data cannot silently alter the claim being tested.

TABLE 3.2: Planned evidence tiers and corresponding limits on inference.

| **Tier** | **Source** | **Authorized Use** | **Prohibited or Unsupported Inference** |
|---|---|---|---|
| 1 | Consenting Bangladesh business invoice, stock, purchase, batch, expiry, customer, and constraint records | Primary business forecasting, customer-risk, strategy, and usability evaluation | Generalization beyond sampled businesses or untested transfer across SME verticals |
| 2a | Bangladesh market-price data, including relevant WFP/DAM sources [25] | Local external context and separate market-price analysis | Direct inference of SKU-level business sales, customer behaviour, or business revenue when those variables are not observed |
| 2b | UCI Online Retail II data from the United Kingdom [3] | Real-invoice pipeline testing and portability checks for selected methods | Claims about Bangladeshi purchasing behaviour or local business effectiveness |
| 2c | Public Bangladesh retailer demand-series data [14] | Local quantity-forecast procedure and implementation checks | Multi-SKU, customer, price, inventory, or profit claims when the required fields are absent |
| 3 | Generated or synthetic records | Software testing, rare-case testing, scalability checks, and privacy-safe examples | Claims of field effectiveness, real customer behaviour, or local operational impact |

Each incoming dataset will be accompanied by a manifest recording its source, owner or repository, collection period, receipt date, checksum, licence or agreement, row count, schema version, mapping rules, exclusions, and known limitations.

Tier-3 generated data will never be mixed into a Tier-1 test set. If generated or augmented records are used during model development, results from real-only training and augmented training will be reported separately and evaluated against the same untouched real test period wherever the required real evidence is available.

### 3.2.7 Data Acquisition and Variable Availability

Primary data acquisition will preferentially use exports from the business's existing point-of-sale, accounting, inventory, or spreadsheet workflow. Where no consistent export mechanism exists, a controlled collection template will be provided.

The minimum sales-related fields are expected to include organization, timestamp, branch, invoice identifier, SKU, quantity, unit price, discount, and cancellation or return status. Recommended additional fields include unit cost, promotion status, payment mode, pseudonymous customer identifier, stock on hand, stock movement, purchase receipt, supplier lead time, and incoming quantity.

Sector-specific extensions capture product category, brand, unit conversion, lot and shelf life where relevant for inventory businesses. Service businesses may instead provide service type, booking and completion timestamps, resource availability and capacity. Non-applicable fields remain explicitly not applicable.

Owner-maintained constraint inputs may include purchasing budget, storage capacity, minimum-order quantity or pack multiple, category-specific limits, and risk preferences. Such constraints shall carry an effective date wherever their historical value can change.

Variables are classified according to their availability relative to the decision cutoff.

TABLE 3.3: Feature availability classes and corresponding leakage rules.

| **Availability Class** | **Examples** | **Rule** |
|---|---|---|
| Known before cutoff | Historical sales, recorded current stock, observed price, known lead time, and calendar information | Eligible when timestamp, provenance, and availability at the historical decision time are valid |
| Known for the forecast horizon | Confirmed promotion, planned closure, or a calendar event known in advance | Eligible only when the same information would genuinely have been available in operational use |
| Observed after the outcome | Later purchase, realized customer response, realized return, or future stock adjustment | May be used as a label or evaluation outcome but never as a predictor for the earlier decision |
| Uncertain or subsequently revised | Unconfirmed promotion, estimated cost, or manually estimated lead time | Retain the source and uncertainty status and evaluate sensitivity where the variable can materially change the action |

### 3.2.8 Data-Quality Assessment and Task Eligibility

Data-quality assessment will be performed before feature engineering and repeated after each substantial structural transformation. A single dataset-wide average will not be used as the sole quality claim because an apparently high overall score may conceal failure in a field that is essential to a particular analytical task. Instead, quality profiles will be evaluated by organization, branch, field, period, and analytical task where applicable.

For required field $j$ within an eligible row set $G$, completeness is defined as

$$C_{j,G} = 1 - \frac{\sum_{i \in G} \mathbb{I}(x_{ij} \text{ is missing})}{|G|}. \tag{3.1}$$

Completeness does not imply validity. Product identifiers will therefore be tested for referential integrity; invoice-line totals will be recalculated from quantity, price, and discount; returns and cancellations will be reconciled with their originating transaction when the relevant identifier is available; and payments will be checked against invoice settlement status.

Duplicate detection will distinguish exact repeated records from legitimate recurrent purchases. Candidate duplicates will be recorded and resolved through a declared rule rather than removed merely because their values appear similar.

Inventory records provide an additional structural reconciliation test. Let $I_{p,t}^{\text{open}}$ and $I_{p,t}^{\text{close}}$ denote the recorded opening and closing inventory balances for product $p$ during interval $t$, and let $\Delta I_{p,t,m}$ denote a signed inventory movement $m$, such as a receipt, sale, return, damage event, or manual correction. The reconciliation residual is

$$r_{p,t} = I_{p,t}^{\text{close}} - I_{p,t}^{\text{open}} - \sum_{m} \Delta I_{p,t,m}. \tag{3.2}$$

The residual is expected to be zero when balances and all relevant movements are recorded consistently. A nonzero residual will not be forced to zero. Instead, its magnitude, duration, and probable cause will be recorded as a data-quality event, and affected observations will be flagged as sensitive or excluded from analyses for which the inconsistency is material.

Temporal-quality tests will also identify implausible event sequences, such as stock receipt recorded before its source purchase order, negative or implausible lead times, customer attributes used outside their valid period, or product history appearing after the analytical cutoff. Event time and ingestion time will be stored separately where available. When only a date is known, the system will not invent artificial within-day temporal precision.

Task eligibility will then be determined from the evidence required by each analysis rather than from a single arbitrary missing-data threshold. Demand forecasting requires a sufficiently interpretable product-time series and a defensible definition of availability. Margin-related KPIs require documented cost provenance. Customer-inactivity modelling requires stable pseudonymous identity and sufficient future follow-up. Replenishment evaluation additionally requires recorded or defensibly reconstructed inventory, lead-time assumptions, relevant operational constraints, and cost ranges. Customer-retention actions require an appropriate contact and consent mechanism.

Any business–task–period combination that violates a required condition will be declared ineligible for that analysis, and the reason will be included in the audit record.

This eligibility gate prevents statistical imputation from manufacturing evidence that the business never recorded. Imputation may still be used for a model feature when its missingness mechanism and decision-time availability are understood, the imputation transformer is fitted using training data only, and an ablation or sensitivity analysis evaluates whether the choice materially affects the result. Imputation shall not be used to

fabricate identifiers, customer consent, future outcomes, operational constraints, or the ground truth of the primary test.

Figure 3.2 turns these rules into an explicit routing decision. Records that fail a repairable structural check return to controlled mapping; records that lack indispensable evidence remain quarantined or make only the affected task ineligible.

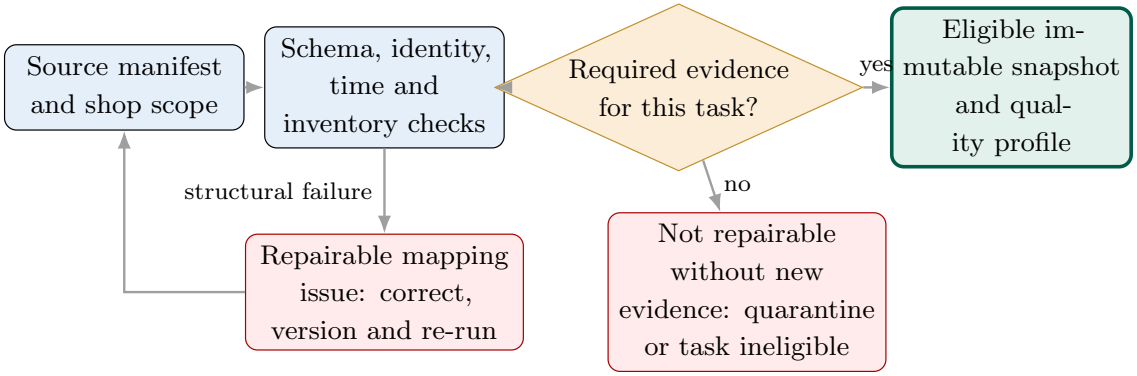



FIGURE 3.2: Data-quality and task-eligibility gate before analytical snapshot release.

### 3.2.9 Ethics, Privacy, and Governance

University ethics approval and an appropriate signed data-sharing agreement will be obtained before primary data collection begins. Direct identifiers such as customer names, phone numbers, email addresses, national identifiers, account numbers, or mobile-wallet numbers are not required for the research analytical tables.

Where customer-level analysis is permitted, each participating business will use a separate salted pseudonymous customer key. Any mapping capable of re-identification will remain under the control of the participating business and will not be stored in the research repository.

The data-sharing agreement will specify the research purpose, authorized researchers, storage arrangements, retention period, publication format, withdrawal conditions, and deletion procedures. Access to research data will follow least-privilege principles, while published outputs will consist of aggregated results or individually reviewed de-identified examples.

A model-generated customer-retention recommendation does not itself authorize contact with that customer. Customer consent and the participating business's contact policy remain independent feasibility constraints. Similarly, system outputs are intended to support rather than replace the decision-making authority of the owner.

### 3.2.10 Canonical Operational and Analytical Model

The proposed operational schema includes the principal entities and events required to preserve business context and analytical traceability. These include `Organization`, `Branch`,

`Product`, `SectorProductProfile`, `InventoryLot`, `ConstraintProfile`, `Customer`, `Supplier`, `SalesOrder`, `SalesOrderItem`, `Payment`, `InventoryBalance`, `StockMovement`, `PurchaseOrder`, `Expense`, `LedgerEntry`, `SalesReturn`, `Refund`, `ImportBatch`, and `AuditLog` objects.

The analytical layer will not train models directly from mutable operational tables. Instead, it will construct versioned invoice-line, SKU-time, and customer-cutoff analytical views that retain source identifiers, transformation metadata, and the relevant historical data cutoff. This separation is intended to preserve reproducibility even when operational records are later corrected or updated.

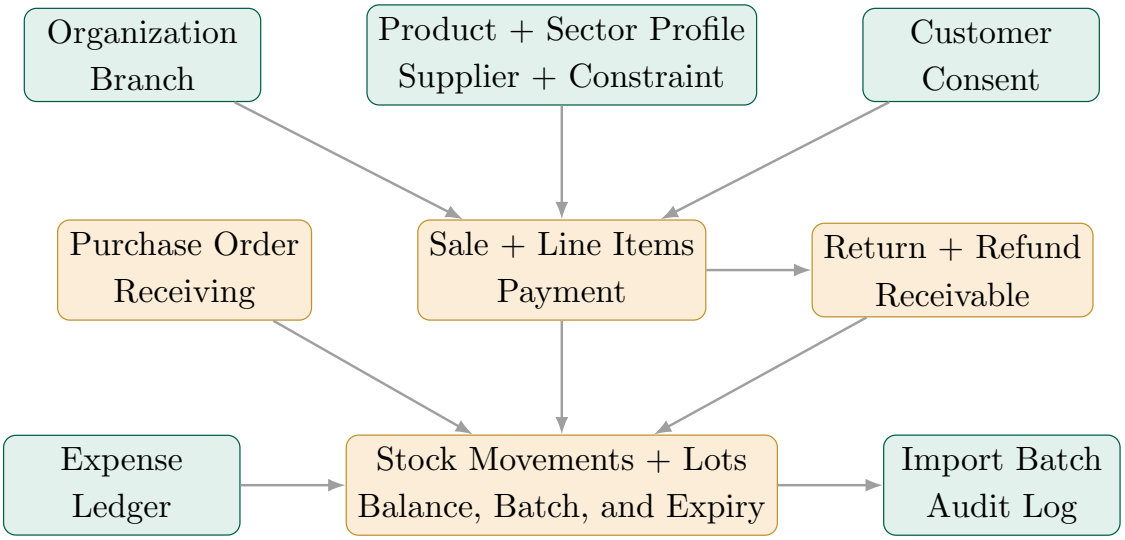



FIGURE 3.3: Proposed canonical operational data model.

Validation rules will examine the uniqueness of business-scoped identifiers, chronological consistency, nonnegative physical quantities where appropriate, payment and order reconciliation, cancellation handling, pack conversion, batch and expiry consistency, inventory balances, and stock movements.

Corrections will be represented as documented and reproducible transformations rather than silently rewriting the historical source evidence. This enables the analytical layer to retain a defensible relationship between operational events, historical snapshots, analytical features, model outputs, recommendations, and subsequent evaluation.

## 3.3 Formal Problem Formulation

Let $b$ denote a business, $l$ a branch, $p$ a product, $r$ an inventory batch, $c$ a customer, and $t$ a decision time. Let

$$D_t = \left\{ \begin{array}{c} \text{Sales, Stock, Batches, Expiry, Purchases, Expenses,} \\ \text{Customers, Constraints, and Context} \end{array} \right\}_{\leq t}, \tag{3.3}$$

denote the governed evidence available at or before decision time $t$. Accordingly, any analytical output produced at $t$ must be derived only from information contained in $D_t$.

### 3.3.1 Performance Evaluation

For each KPI $K$, the system returns $K(D_t; W, g)$, where $W$ is the reporting window and $g$ is the aggregation level. Each KPI retains its numerator, denominator, filters, definition version, and data-quality status so that the owner can reproduce and inspect the calculation.

For reporting window $W$, net sales are calculated from valid invoice lines using the documented business treatment of discounts, cancellations, and returns. A general form is

$$\text{NetSales}_W = \sum_{i \in W} q_i p_i - \sum_{i \in W} d_i - \sum_{r \in W} v_r, \tag{3.4}$$

where $q_i p_i$ is the gross value of invoice line $i$, $d_i$ is the recognized discount, and $v_r$ is the value of a valid return or reversal attributed to the reporting window.

Gross margin is reported only when sufficiently reliable cost evidence is available:

$$\text{GrossMarginRate}_W = \frac{\text{NetSales}_W - \text{COGS}_W}{\text{NetSales}_W}, \tag{3.5}$$

provided that $\text{NetSales}_W \neq 0$.

If cost information is unavailable for a material share of sold units, the system reports sales and cost coverage rather than silently interpreting revenue as profit. Average order value uses eligible completed orders as its denominator; repeat-rate reporting states customer-identification coverage; and stock-turnover reporting specifies both the cost and average-inventory conventions used. Period comparisons use comparable windows or explicitly identify differences in trading days and closures.

Each KPI response therefore contains a value, definition version, reporting period, aggregation level, coverage indicators, and reconciliation status. This structure makes changes in business-rule definitions visible and allows the evaluation to distinguish calculation anomalies from genuine changes in business performance.

### 3.3.2 Demand Prediction

Demand forecasting is defined at a specific aggregation level. The first implemented experiment forecasts observed units at shop–category–day level; it does not establish SKU-level latent demand. The prepared SKU export contains positive-sale records only. A SKU-date without a row is not automatically a zero-demand day. SKU targets require

verified active assortment, shop operating calendar and stock exposure. Stock-out, negative-ledger, closed-shop, incomplete-inventory and unknown-assortment periods are censored or unknown rather than zero.

For series $s$, origin $t$, and horizon $h \in \{1, 7, 14, 28\}$, the direct cumulative target is

$$Y_{s,t,h} = \sum_{j=0}^{h-1} y_{s,t+j}.$$

A separate model is fitted for each horizon. All predictors must have been available strictly before $t$; future sales, returns, inventory movements and realized customer responses are outcomes, not features. A future promotion is eligible only when its approval or publication timestamp proves that it was known at the forecast origin.

The historical feature family comprises lags 1, 7, 14, 28 and 56; rolling mean, standard deviation, sum and zero-sales ratio over 7, 14, 28 and 56 days; days since last positive sale; ADI; positive-demand mean and $CV^2$; shop and category identifiers; weekday, day, month, ISO week and cyclical calendar encodings. Price, discount, promotion, lead-time, stock and supplier features are added only after their point-in-time availability and joins have been validated.

The candidate ladder includes weekly seasonal-naive, Croston–SBA, TSB, pooled LightGBM Tweedie, CatBoost Tweedie and a two-stage occurrence– quantity (hurdle) candidate. Croston-family methods model demand size and inter-demand intervals [6, 24]; TSB updates occurrence probability over time [27]. LightGBM is a gradient-boosted tree candidate [26]; CatBoost is included for categorical identifiers [29]. The hurdle candidate estimates

$$\widehat{E(Y \mid X)} = \widehat{P(Y > 0 \mid X)}\, \widehat{E(Y \mid Y > 0, X)}.$$

Its probabilities must be calibrated before the model is considered for operational selection [28].

Demand regimes are described using ADI and $CV^2$, with descriptive thresholds 1.32 and 0.49. Regime labels are calculated from history available at the forecast origin. They stratify error and support candidate diagnosis; they do not automatically assign a model. A specialist can be promoted only after same-origin validation shows stable improvement and there are at least three development origins with at least twenty independent series in the regime at each origin. If support is insufficient, the pooled fallback is retained.

The initial fixed-origin development experiment found LightGBM to be the strongest candidate at 7-, 14- and 28-day shop–category horizons. CatBoost remains a challenger. The raw hurdle produced a lower one-day WAPE but material under-forecasting and uncalibrated probabilities. These results are controlled development evidence because the supplied package's provenance is unresolved; they do not establish real-world SME performance.

Forecast uncertainty will be assessed with calibrated quantile or residual intervals. SKU, category, branch and business forecasts may be reconciled only after source coverage and aggregation identities are verified [12]. The forecast estimates observed sales, not unconstrained latent demand, unless stock-out censoring has been adequately measured.

### 3.3.3 Adaptive Forecasting Architecture and Promotion Gates

The proposed architecture separates evidence eligibility, forecast generation, model selection and business action. Figure 3.4 shows the sequence. The source gate blocks unverified data from real-world claims; the target builder distinguishes observed zero, censored and unknown demand; the candidate layer fits transparent baselines and eligible specialists; and the selection gate uses common rolling origins, bias and minimum independent-series support. A decision layer may use a selected forecast only after inventory, budget, lead-time and owner constraints have been validated.

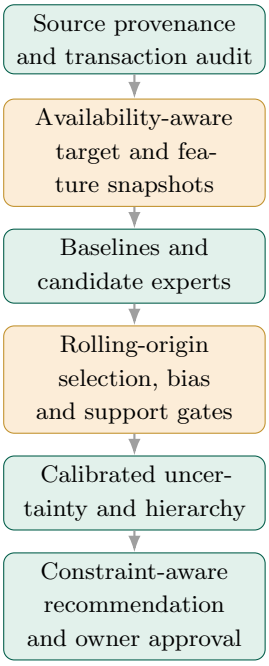



FIGURE 3.4: Evidence-gated adaptive demand forecasting architecture.

### 3.3.4 Future Customer Inactivity

For customer $c$ at historical cutoff $t$, future inactivity is represented as

$$q_{c,t} = P(\text{no purchase in } (t, t+H] \mid X_{c,t}), \tag{3.6}$$

where $X_{c,t}$ contains only records available by cutoff $t$, and $H$ is the predefined future outcome horizon. The value of $H$ will be fixed before final evaluation with reference to the relevant business purchase cycle.

The resulting $q_{c,t}$ represents predicted inactivity risk and does not, by itself, estimate the causal effect of contacting customer $c$.

### 3.3.5 Strategy Optimization

For an eligible action $a \in A_t$, the system evaluates

$$U(a \mid D_t) = \mathbb{E}[\Delta\text{Profit}(a)] - \text{ActionCost}(a) - \lambda\,\text{Risk}(a), \tag{3.7}$$

subject to operational constraints including budget, storage, minimum-order quantity, supplier lead time, pack multiple, remaining shelf life, expiry, temperature-controlled storage capacity, incoming stock, available inventory, and customer-contact consent where applicable.

This transparent objective represents the transition from predictive outputs to prescriptive decisions. Its terms will be estimated using available evidence, and uncertain values will be examined through sensitivity analysis rather than treated as precisely known constants.

#### 3.3.5.1 Inventory Position and Replenishment Cost

For replenishment, the inventory position immediately before an action is

$$\text{IP}_{p,t} = \text{OnHand}_{p,t} + \text{OnOrder}_{p,t} - \text{Backorder}_{p,t}. \tag{3.8}$$

Let $D_{p,t}^{(L)}$ denote uncertain demand for product $p$ over the supplier lead-time period $L$, and let $Q$ be a candidate order quantity. A transparent conditional replenishment-cost function is

$$J_p(Q) = c_o\,\mathbb{I}(Q > 0) + c_h\,\mathbb{E}\left[\left(\text{IP}_{p,t} + Q - D_{p,t}^{(L)}\right)^+\right] + c_s\,\mathbb{E}\left[\left(D_{p,t}^{(L)} - \text{IP}_{p,t} - Q\right)^+\right], \tag{3.9}$$

where

$$x^+ = \max(x, 0), \tag{3.10}$$

and $c_o$, $c_h$, and $c_s$ denote declared ordering, holding, and shortage costs, respectively.

The feasible set additionally imposes supplier pack size, minimum-order requirements, storage limits, category-level purchasing limits, and available cash. Equation 3.9 is an evaluation model and does not imply that these costs are known precisely. Candidate

policies will therefore be compared across owner-reviewed parameter ranges. A specific savings claim will not be reported when the ranking of candidate policies changes materially across plausible cost values.

Uncertainty is represented through the lead-time demand distribution rather than through a single inflated point forecast. Candidate order quantities are also screened against pack conversion, batch availability, remaining shelf life, expiry, and storage conditions. Consequently, high predicted demand cannot justify an operationally prohibited action.

Where reliable predictive quantiles are available, candidate quantities may be compared with those quantiles. Otherwise, residual- or replay-based scenarios from validation periods will be used. If stock-out censoring prevents credible reconstruction of unconstrained demand, the result will be reported as a policy comparison under observed sales rather than as an estimate of fully unconstrained demand.

#### 3.3.5.2 Customer Action Logic and Abstention

Customer-action logic differs from replenishment logic. The inactivity probability $q_{c,t}$ in Equation 3.6 may be used to prioritize customers for review, but it does not estimate the causal effect of contacting customer $c$.

A contact action is eligible only when customer consent and business policy permit contact, the proposed treatment is specified, and sufficient operational capacity exists. Expected profit will be calculated only when a defensible estimate of incremental response benefit is available. Otherwise, the output will explicitly remain a risk-ranked contact list. This separation prevents a predictive probability from being incorrectly interpreted as campaign uplift.

The action layer may also abstain from making a recommendation. Abstention is appropriate when required constraints are missing, uncertainty makes the leading alternatives practically indistinguishable, predicted benefit does not exceed action cost over the declared sensitivity range, or the relevant model fails a monitoring gate. An abstention accompanied by a reason is therefore treated as valid safety behaviour rather than as a system error.

### 3.3.6 Preprocessing and Historical Snapshots

The processing sequence is fixed: validate the source and schema; remove or flag exact duplicates; reconcile cancellations and returns; construct the data split; fit imputation, encoding, and scaling transformations using training data only; construct features; train candidate models; and evaluate them on the untouched test set. This ordering prevents information from future periods from leaking into model development [15].

Demand features may include lagged demand, rolling location and variability statistics, day of week, month, distance to relevant festivals, price, discount, known promotions, stock-out indicators, pack units, remaining shelf life, and external market context. Batch-expiry and

temperature-controlled storage variables primarily constrain action feasibility and enter prediction only when their decision-time availability and analytical role are defensible.

Where feasible, zero observed sales caused by zero demand will be distinguished from zero sales caused by lack of stock. Intermittency will be measured before selecting temporal granularity and candidate forecasting methods.

Customer snapshots may include RFM variables, tenure, order-value statistics, category breadth, purchase intervals, discount behaviour, and pre-cutoff return rates. The outcome label is constructed from a subsequent future window. Where required, a purge interval will separate the observation and label windows to prevent overlap or delayed events from introducing leakage.

Customer identifiers used for an entity-generalization test will not appear on both sides of the corresponding training–test boundary.

## 3.4 Candidate Methods and Selection Rules

### 3.4.1 Baseline Ladder and Task-Specific Candidates

Method selection follows a baseline ladder, with model eligibility determined by the available evidence volume and task structure.

- **Performance:** Auditable KPI definitions and rule-based anomaly detection provide the descriptive baseline.
- **Demand:** Last-value, weekly seasonal-naive, moving-average, and exponential-smoothing baselines are evaluated before ARIMA/SARIMA, Croston–SBA, TSB, and gradient-boosted lag models. LightGBM and CatBoost are candidates for pooled shop/category or SKU features when the sample and categorical structure support them. LSTM models are considered only when sequence volume and history justify their additional complexity. Intermittent-demand methods are promoted only after same-origin validation [6, 24, 26, 27].
- **Inactivity:** A recency or base-rate rule is compared with Logistic Regression, Random Forest [2], and XGBoost [5]. Class weighting and decision thresholds are selected using development data only.
- **Segmentation:** Appropriately scaled RFM features are analysed using K-Means. Candidate values of $K$ are evaluated using inertia, silhouette, resampling stability, and business interpretability.
- **Explanation:** Coefficient- or rule-based explanations are preferred for simple models, while SHAP provides model-behaviour attributions for compatible tree-based models [16]. Displayed explanations will remain selective and context-specific [18].

### 3.4.2 Model Eligibility and Sector-Aware SME Adaptation

A complex candidate is rejected when the training history is insufficient, when it fails to improve on the relevant baseline across common development origins, when its probabilities or residuals are unstable, or when its operational cost has no material value. For the current demand task, a regime-specific selector requires at least three development origins and at least twenty independent series in that regime at each origin. Repeated forecasts of the same series do not count as independent support. If a regime fails this gate, the pooled fallback remains selected and the regime-specific result is reported as insufficiently supported.

Model eligibility also follows the available sector-aware SME evidence. A new or data-limited business uses a transparent baseline and may use a pooled model trained on authorized, de-identified evidence from comparable businesses with sector and operating-model context. With moderate history, shop- and branch-specific context may calibrate the pooled model. A fully shop-adapted candidate is considered only when sufficient local history exists.

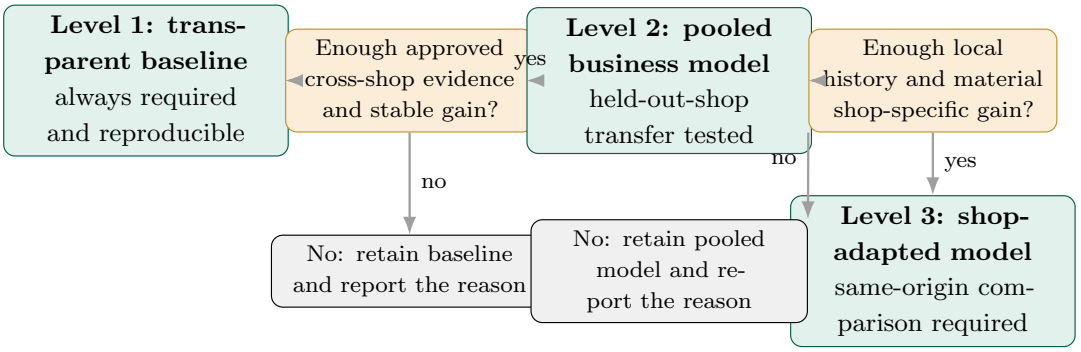



FIGURE 3.5: Evidence-dependent model promotion ladder for each business analytical task.

## 3.5 The B-SMART Sequential Procedure

### 3.5.1 Strategy Recommendation Algorithm

**Algorithm 1:** B-SMART Strategy Recommendation

**Input:** Data available at cutoff $D_t$, horizon $H$, constraints $C_t$, validated model/baseline versions $M$, and candidate actions $A$

**Output:** Ranked feasible recommendations $R_t$

Check sources, schema, timestamps, and organizational ownership;
Create the historical feature snapshot $X_t$;
Apply transformations fitted on the appropriate training evidence;
Compare eligible baseline, pooled-business, and shop-adapted candidates;
Estimate demand uncertainty and customer inactivity probabilities;
Compute reproducible KPIs and data-quality alerts;
Generate candidate actions with source and decision-time metadata;
**foreach** $a \in A$ **do**
    Check budget, stock, pack, applicable shelf-life, storage, lead-time, capacity, and consent feasibility under $C_t$;
    Estimate benefit, cost, and uncertainty using declared assumptions;
    $U(a) \leftarrow \mathbb{E}[\Delta\text{Profit}(a)] - \text{Cost}(a) - \lambda\,\text{Risk}(a)$;
Remove infeasible actions and rank the remaining actions by $U(a)$;
Attach a concise explanation, uncertainty statement, cutoff, and evidence source;
Return the top-$K$ actions and record the owner's response and any subsequent observable outcome;

### 3.5.2 Computational Complexity

For $N$ invoice lines, $d$ validated fields, and $P$ candidate actions, source validation and feature aggregation are approximately

$$O(Nd), \tag{3.11}$$

while ranking the candidate actions requires approximately

$$O(P \log P). \tag{3.12}$$

The computational complexity of model training depends on the selected model and will be measured empirically. Serving uses versioned artifacts and therefore does not retrain the model within the request path.

## 3.6 Proposed Architecture and Information Flow

### 3.6.1 Architecture Components

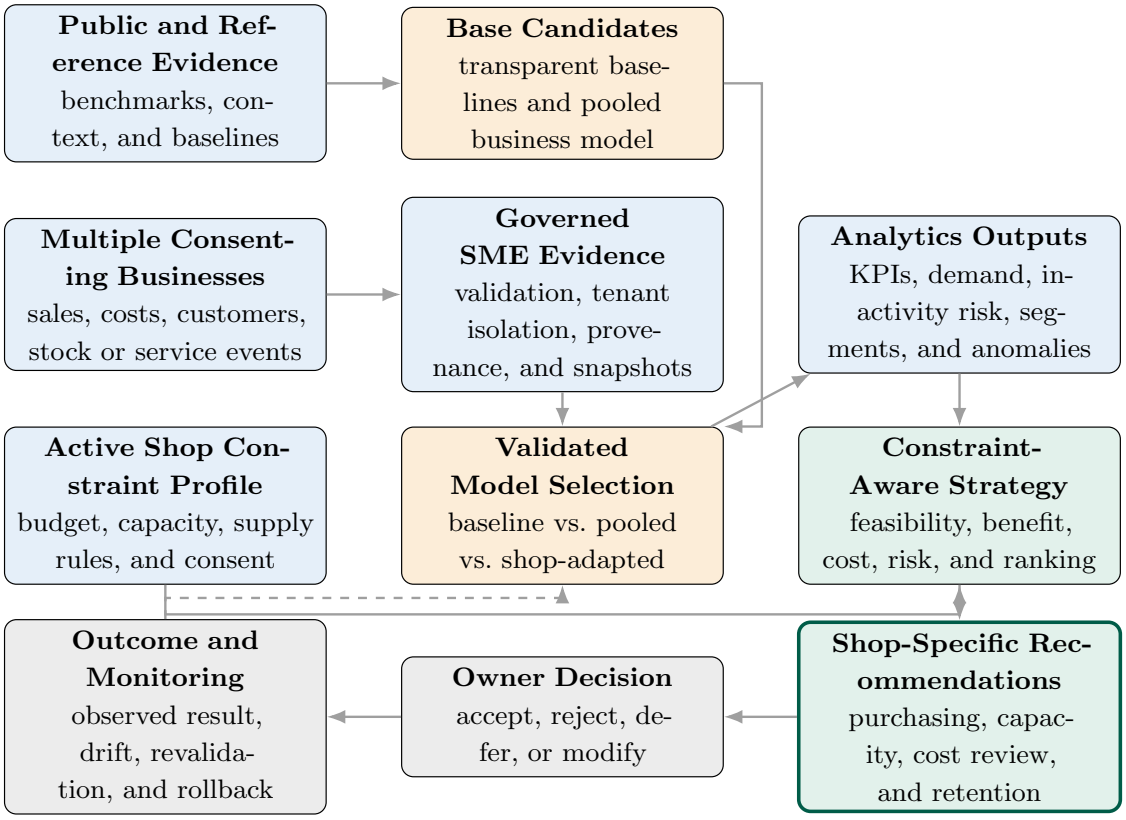



FIGURE 3.6: SME-specific constraint-aware B-SMART architecture based on the supervisor-guided flow.

The architecture separates the pooled business-level model from shop-level personalization. Authorized and de-identified records from multiple businesses may support a pooled business model, while each deployed business retains its isolated operational data and active constraint profile.

The selected analytical model produces estimates rather than explicit orders. The strategy layer combines those estimates with the shop's budget, inventory, pack, applicable shelf-life, storage, lead-time, capacity, and consent constraints to generate feasible recommendations.

### 3.6.2 Monitoring and Continuous Learning

Information flow is deliberately one-way at prediction time: future outcomes cannot alter the historical snapshot that generated an earlier prediction or recommendation.

Operational state and recommendations may be updated when relevant events occur or according to a defined schedule. However, the deployed model is not blindly retrained after every transaction.

Observed outcomes enter the monitoring process and become eligible training evidence only in a later model version after reconciliation, label maturation, drift assessment, and comparison with the existing model and baseline.

This monitored feedback lifecycle represents the intended meaning of continuous learning while preserving the ability to reproduce what the system knew when a particular action was proposed.

## 3.7 Snapshot Lineage and Frozen Evaluation Flow

### 3.7.1 Historical Snapshot Construction

The analytical snapshot is the unit used to prevent future information from entering an earlier decision. Figure 3.7 distinguishes the observation period, decision cutoff, and subsequent outcome window. Raw events are first retained in a governed and versioned event ledger. The snapshot builder then selects only records available by cutoff $t$, applies declared mappings, and constructs task-specific features from evidence that was available at that time.

The resulting snapshot is immutable for the corresponding analytical release. Its required attributes include the source-event references, decision cutoff, mapping and feature versions, applicable eligibility rules, and the model or baseline version used to generate the released output.

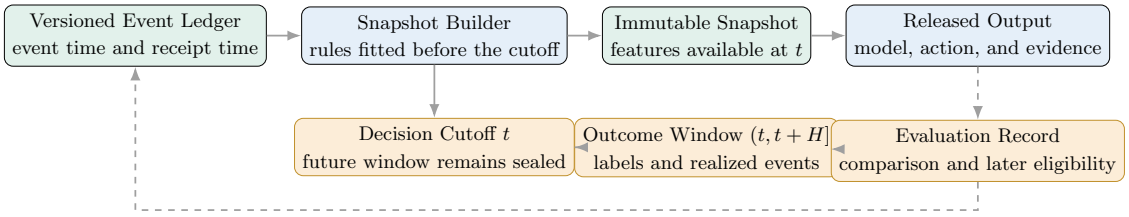



FIGURE 3.7: Historical snapshot construction and separation of future outcomes.

For forecasting, multiple historical cutoffs form rolling forecast origins. Transformations such as scalers, encoders, imputers, and feature-selection rules are fitted only within the relevant training window and then applied to the corresponding validation origin.

For customer analysis, each snapshot contains behaviour observed only up to cutoff $t$. A snapshot receives an inactivity label only when the complete future horizon $H$ can be observed. Where delayed returns, fulfilment events, or rolling features could cross the boundary, a purge interval is inserted. For entity-generalization analysis, customer or product grouping is performed before any learning transformation.

### 3.7.2 Frozen Final Evaluation and Outcome Maturation

Training and validation snapshots are used for model development. The final test set remains sealed until the task definition, feature and eligibility rules, candidate models, primary metrics, decision thresholds, and random seeds have been frozen.

If a defect is identified after the final test has been opened, the defect and its impact will be documented. The affected observations cannot subsequently be treated as though they were still part of an untouched final test. This rule protects the interpretation of the final result rather than merely enforcing the mechanics of a data split.

Outcome records are not automatically incorporated into a released model. Observed outcomes first enter the monitoring store. They become eligible for a later model version only after reconciliation, label maturation, and repetition of the appropriate validation protocol.

This delay ensures that an owner's response, a corrected invoice, or a partially observed outcome cannot retrospectively change the evidence that supported a historical production decision.

## 3.8 Owner-Facing Recommendation Evidence Contract

### 3.8.1 Minimum Recommendation Evidence

Every displayed recommendation will satisfy a minimum evidence contract. The contract separates the proposed action from the evidence required to judge that action. Figure 3.8 summarizes the required components.

The interface may initially display only the most important information, but the complete recommendation record must be retained for audit and evaluation.

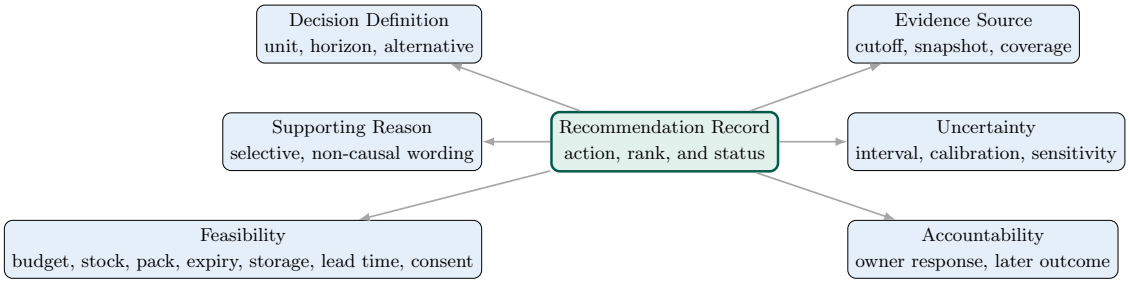



FIGURE 3.8: Minimum evidence contract for an owner-facing recommendation.

The action component specifies precisely what is proposed and which comparator would otherwise be used. The evidence component records the cutoff, source, snapshot, and coverage so that stale or incomplete data can be identified.

The reason component uses a small number of specific supporting factors and avoids causal wording when only associative evidence is available. The uncertainty component reports intervals, calibration information, or sensitivity where applicable and helps determine whether nearby alternatives are practically different.

The feasibility component documents the operational constraints applied to the recommendation. Finally, the accountability component records whether the owner accepted, rejected, deferred, or modified the recommendation, together with an optional reason and any later observable outcome.

### 3.8.2 Human Use and Accountability

An explanation is considered successful only when it contributes to correct and appropriate use. The owner study will therefore include scenarios in which the appropriate decision is to reject or defer an apparently attractive recommendation.

This evaluates calibrated human reliance rather than merely whether participants prefer a more detailed interface. The evidence contract also supports failure analysis because an inappropriate decision can be traced to the underlying estimate, missing constraint, explanation, outdated evidence, or owner interpretation rather than being attributed generically to the "AI."

## 3.9 Evaluation Protocol

Figure 3.9 summarizes the evaluation sequence shared by the forecasting, customer and decision studies. The locked final test is opened only after preprocessing, candidates, thresholds, primary metrics and decision rules have been frozen; any later revision is reported as exploratory.

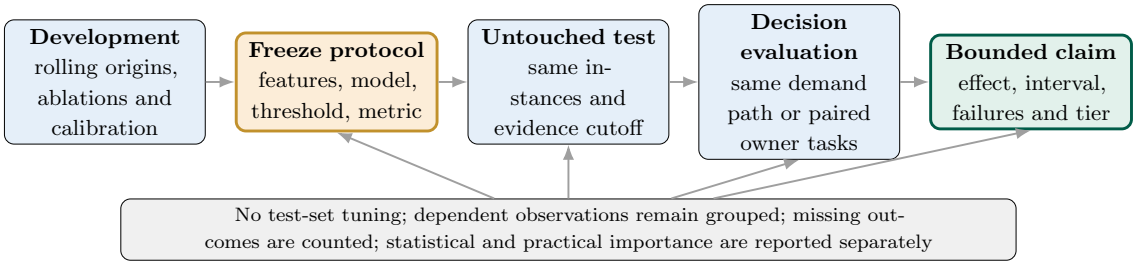



FIGURE 3.9: Frozen evaluation sequence from model development to a proportionate claim.

### 3.9.1 Forecasting Evaluation

All forecasting splits preserve temporal order. Candidate methods are selected using rolling-origin or expanding-window validation, followed by one evaluation on a final untouched period.

For observed values $y_i$ and forecasts $\hat{y}_i$, the principal error measures include

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|\,, \tag{3.13}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}, \tag{3.14}$$

$$\mathrm{WAPE} = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{\sum_{i=1}^{n} |y_i|}, \qquad \sum_{i=1}^{n} |y_i| > 0. \tag{3.15}$$

MASE will additionally support scale-independent comparison and avoids several limitations of conventional percentage errors [13]. For the current cumulative-horizon benchmark, the reported horizon-scaled weekly MASE uses the per-series mean absolute lag-7 difference multiplied by the forecast horizon. Cluster bootstrap resampling is performed at the shop–category-series level to preserve repeated-origin structure. Prediction-interval coverage and interval width will be used to evaluate uncertainty.

Results will be stratified by demand volume and intermittency so that poor performance on low-volume or sparse products is not hidden inside an aggregate average.

### 3.9.2 Customer and Segmentation Evaluation

Customer-inactivity models will report outcome prevalence, PR-AUC, ROC-AUC, Lift@10%, Brier score, and probability-reliability analysis. Precision–recall measures are emphasized under class imbalance [21], while calibration is evaluated independently because a model may produce useful rankings while still producing misleading probabilities [19].

Decision thresholds and contact fractions will be selected using validation evidence and declared cost assumptions. They will not be modified after the final test outcomes have been inspected.

Clustering evaluation will report inertia, silhouette, cluster stability, minimum cluster size, and corresponding RFM profiles. Final segment interpretations will also be reviewed with domain participants.

If separation is weak or unstable, the result will be reported as such. The existence of algorithmically generated clusters will not be treated as evidence that natural or causal customer types exist.

### 3.9.3 Ablation, Robustness, and Statistical Analysis

Planned ablations include local-calendar features, price and promotion features, market context, pack and expiry context, and customer-history extensions.

Comparisons will include the transparent baseline, eligible pooled-business models, and shop-adapted models where the available evidence permits them. Where augmentation is used, real-only and augmented variants will be evaluated under controlled comparisons.

Results will be presented by business, with paired comparisons conducted over common forecast origins or comparable analytical instances. A held-out-business transfer test will be used where sufficient participating businesses are available.

Uncertainty will be summarized using paired block-bootstrap intervals across time or business units, depending on the relevant dependence structure. Differences between models will be reported using effect estimates and uncertainty intervals rather than significance labels alone.

Sensitivity analysis will vary supplier lead time, shortage cost, holding cost, action cost, and the action-risk penalty used in Equation 3.7.

### 3.9.4 Frozen Analysis and Statistical Decision Rules

One primary comparison and one primary metric for each research question will be frozen before the final test is opened.

For RQ1, the primary comparison is based on paired out-of-time forecasting errors at common forecast origins. WAPE or MASE will be selected according to denominator validity and scale suitability.

For RQ2, PR-AUC is the primary discrimination measure, while probability calibration is treated as a separate acceptance criterion.

For RQ3, competing replenishment policies are compared on the same demand paths, with the same opening inventory and the same applicable constraints.

For RQ4, task correctness is the principal owner-facing outcome, while completion time and confidence provide complementary evidence regarding the practical effect of the explanation condition.

The unit used for uncertainty estimation will match the relevant dependence structure. Forecast errors from neighbouring dates will not be treated as independent observations. Block-bootstrap resampling will preserve local temporal dependence.

When multiple businesses are available, between-shop differences will be reported rather than hidden within a pooled transaction count. A leave-one-business-out or held-out-business analysis will assess whether a pooled business model transfers to an unseen shop.

Customer comparisons will account for repeated snapshots from the same customers. Usability-task comparisons will be paired within participants where the study design permits. Effect estimates will be accompanied by confidence intervals so that the remaining uncertainty is visible.

### 3.9.5 Practical Importance and Missing Outcomes

Practical importance will be evaluated independently of statistical evidence. A small reduction in average forecasting error may be operationally important for a high-volume product but irrelevant after ordering constraints are applied. Similarly, a moderate increase in task-completion time may be acceptable if it produces a material improvement in decision correctness.

Where defensible cost or effect thresholds can be established with participating businesses, they will be specified before the final comparison. Where no defensible threshold exists, the magnitude and uncertainty of the effect will be reported without assigning an arbitrary success label.

Secondary metrics, alternative horizons, customer segments, and feature interactions remain useful for diagnosis but will be identified as secondary or exploratory analyses. When many related comparisons are examined, the full set of relevant results will be reported rather than only favourable cases.

Any post-test change to a model, threshold, feature definition, or analysis creates a new exploratory version rather than replacing the original primary result. This separation allows continued model improvement without rewriting the original evidence.

Missing outcomes will be explicitly identified. An incomplete future horizon cannot automatically be treated as evidence of customer inactivity. An unknown inventory balance cannot be treated as zero stock. Similarly, a recommendation without a later observable outcome may support a usability analysis but cannot support a claim of realized business impact.

Where assumptions can reasonably be bounded, sensitivity analysis will be used. Otherwise, the corresponding claim will be classified as not evaluable with the available evidence.

### 3.9.6 Decision, Usability, and Scalability Evaluation

Replenishment-policy comparisons will use identical observed or replayed demand paths. Outcomes will include stock-out units, average inventory, holding cost, shortage cost, and total cost.

Retention ranking will incorporate expected value only when a defensible incremental response benefit, contact cost, and customer-consent condition are available. Otherwise, the evaluation remains a prioritization study rather than a realized-profit analysis.

Owner evaluation will use realistic decision tasks within a counterbalanced repeated-measures or within-participant design where appropriate for the available sample. Measures will include task correctness, completion time, confidence, perceived usability, and error count.

Participants may also be asked to explain the reason for their decisions so that the study can identify plausible but incorrect interpretations of the recommendation or explanation.

Generated data will be used for scalability testing and will remain separate from real-data accuracy claims. Planned transaction-line sizes are

$$100, \quad 500, \quad 1{,}000, \quad 10{,}000, \quad 100{,}000, \quad 1{,}000{,}000.$$

Scalability measures will include import time, preprocessing time, query latency, and representative serving latency. Any experiment concerning association rules will be reported only if the corresponding algorithm has actually been implemented and executed.

### 3.9.7 Monitoring Plan

Monitoring will distinguish among schema or data-quality failure, input distribution shift, calibration change, and outcome-performance deterioration.

An alert does not automatically authorize retraining. Any replacement model must pass the validation procedure again and retain its training window, feature version, parameters, and comparison results.

This requirement ensures that adaptation is evaluated before deployment rather than assumed to be beneficial merely because new data have become available.

## 3.10 Reproducibility, Audit, and Failure Reporting

### 3.10.1 Reproducibility Levels

Four levels of reproducibility will be designed.

- **Data reproducibility:** requires a source manifest, checksum, schema version, mapping rules, and exclusion log.
- **Analytical reproducibility:** requires snapshot identifiers, fitted transformations, feature definitions, target construction, and split manifests.

- **Model reproducibility:** requires code, execution environment, parameters, random seeds, trained artifacts, and the evaluation command.
- **Decision reproducibility:** additionally requires the applicable constraints, cost assumptions, candidate-action set, ranking rule, and interface version.

### 3.10.2 Restricted and Shareable Evidence

Primary partner data do not need to be publicly released in order for the study to maintain reproducibility controls. The reproducibility package will therefore distinguish restricted evidence from shareable artifacts.

Shareable materials may include schemas, data dictionaries, manifest templates, transformation code, generated test fixtures, and configuration files without revealing confidential identifiers or transaction records.

Where restricted evidence is available to an authorized evaluator, the available package should permit that evaluator to reconstruct the relevant snapshot and result. An external reader should additionally be able to inspect the process using a public or generated test fixture.

### 3.10.3 Experiment Records and Correction History

Each experiment will generate a machine-readable run record containing, where applicable, the research question, dataset and snapshot hashes, training and test windows, grouping keys, feature version, candidate and baseline versions, parameters, output-artifact hashes, warnings, and start and finish times.

Manual spreadsheet corrections will not be made outside this record. If an owner supplies a correction, it becomes a dated source or mapping revision. The correction affects downstream outputs only when a new version is executed.

### 3.10.4 Failure Reporting

Failure reporting is part of the scientific output. Data-eligibility failure, lack of convergence, numerical instability, training failure, resource exhaustion, and constraint infeasibility will be distinguished from ordinary evaluation outcomes.

The number of attempted cases will be reported. Results will not be summarized only over successful model runs while silently excluding failures.

A method that fails on sparse or short series will therefore be evaluated against the full eligible set to which it was applied, rather than only the subset for which it returned a prediction.

### 3.10.5 Recommendation Auditability and Claim Traceability

Auditability extends to the owner interface. Every recommendation receives an identifier linking the evidence contract, displayed recommendation, owner response, and any later observable outcome.

If a recommendation changes following a data correction or model replacement, the previous record remains immutable and the successor record refers to it. This supports accountability without implying that the model, researcher, or owner alone caused the eventual business outcome.

The final research package will also include a claim-to-evidence index. Each central claim will identify the corresponding research question, figure or table, experimental record, and evidence tier.

Claims supported only by public, generated, simulated, or small-sample evidence will remain explicitly labelled as such. This index provides a final safeguard against expanding the scope of a claim beyond what was actually evaluated.

## 3.11 Threats to Validity and Planned Controls

Table 3.4: Main validity threats and planned controls.

| **Threat** | **Potential Effect on Evidence** | **Planned Control** |
|---|---|---|
| Selection bias | Participating firms may be more digitally mature than the wider target population | Describe recruitment and participating firms explicitly and restrict generalization accordingly |
| Data leakage | Future or target-derived information may artificially improve measured performance | Use availability rules, historical cutoffs, immutable snapshots, and temporal- or entity-safe splits |
| Stock-out censoring | Observed sales may understate unconstrained demand | Retain stock-state information and flag or stratify censored periods |
| Overfitting on small or short series | Complex models may fit noise rather than transferable demand structure | Use model-eligibility rules, simple baselines, and rolling temporal validation |
| Dataset mismatch | Foreign or public benchmark evidence may be incorrectly interpreted as local effectiveness evidence | Use explicit evidence tiers and report results separately by evidence source |
| Weak cost or operational assumptions | Uncertain cost values may materially alter utility rankings | Use owner-reviewed parameter ranges and sensitivity analysis |
| Novelty or learning effect | Interface performance may depend on presentation order or participant learning | Counterbalance task order and collect task-level correctness and timing measures |
| Concept drift | Festival, price, demand, or customer-behaviour relationships may change over time | Use time-based evaluation and post-deployment monitoring with revalidation before model replacement |

## 3.12 Expected Methodological Outputs

At completion, the methodology is expected to produce a documented canonical business schema and data dictionary, a versioned feature and partner-data protocol, and explicit specifications for transparent baselines, pooled-business models, and eligible shop-adapted models.

The study will also produce a reproducible evaluation package, a constraint-aware recommendation procedure, and an owner-study instrument.

Effectiveness claims will be restricted to the sampled businesses, sectors, tasks and periods actually evaluated. Transfer to a new sector requires schema and constraint mapping, new evidence and held-out-sector evaluation.

The system supports business decisions under owner oversight; automated actions outside the evaluated business tasks remain excluded.

Implementation details, observed numerical results, and conclusions will be reported only when the corresponding empirical evidence is available.

# Chapter 4

# Implementation

## 4.1 Dataset Package and Research Scope

The current Drive manifest declares the Bangladesh SME package to contain real SME sales or real people. However, the repository's 2 November 2026 onboarding record classifies a package with the same seed (20261025), entity counts (32 shops, 11,000 customers, 480 products, 56 suppliers), and 2023–2025 coverage as fully synthetic/simulated. The earlier record also reports a `synthetic_flag=YES` field. The currently accessible SKU sheet has six populated columns and does not expose that flag. The declarations therefore conflict.

The evidence class for this report is `provenance_conflict`. Structural reconciliation does not prove real-world origin. Until the source owner supplies traceable POS/accounting exports, collection and consent basis, anonymisation procedure, data dictionary, and a versioned explanation of the earlier synthetic classification, this package is used only for controlled research development and software validation. Its scores are not presented as verified real-world SME accuracy.

The current Drive manifest lists 32 shops, 11,000 customers, 480 products, 56 suppliers and the period 1 January 2023–31 December 2025, with BDT as currency. It lists 365,447 sales orders, 736,683 sales lines, 381,756 payments, 122,422 purchase orders and purchase lines, 994,808 inventory movements, 134,379 inventory lots, 5,102 returns, 1,216 promotions, 1,096 calendar rows, and 32,852 separately identified `simulated_unmet_demand` rows. The latter are not observed sales or empirical lost-sales ground truth and are excluded from all forecast targets.

The RQ1 experiment uses the dedicated `rq1_daily_shop_category` sheet. Its five fields are `shop_id`, `category`, `date`, `units_sold_observed`, and `is_zero_observed_sales`. The resulting panel contains 32 shops, 16 categories, and 512 shop–category series. Each series has 1,096 consecutive daily records, producing 561,152 rows over the stated three-year period. The panel records 1,580,361 observed units.

## 4.2 Data Validation and Preprocessing

The ingestion procedure parses the date field, checks required columns and missing values, rejects duplicate shop–category–date keys, verifies that units sold are finite and non-negative, and checks that the zero-sales flag agrees with the observed quantity. It then verifies that every shop–category series contains the same complete daily calendar. No missing dates are filled by assumption. The audit found zero duplicate keys, zero missing required values, zero zero-flag mismatches, and zero missing dates in the supplied panel. The zero-sales share is 37.5962%.

The dedicated SKU-level export is not substituted for this complete shop–category panel. Its absent SKU-date rows require a separate source-aware decision about assortment, shop opening, product activation, stock-outs, and export completeness before zero demand can be assigned. This distinction prevents the conversion of unobserved records into artificial zero labels.

## 4.3 Transaction and Panel Reconciliation

The 736,683 sales lines join to 365,447 sales orders without missing order references or shop mismatches. Aggregating source sales lines by `shop_id`, `sku_id` and calendar date reproduces all 699,220 rows of the prepared positive-sale SKU table exactly, including 1,580,361 units and the recorded sales value to floating-point precision. This is an internal reconciliation of the supplied files, not an independent audit of their real-world origin.

The prepared SKU export contains 32 shops, 480 SKUs and 13,271 shop–SKU combinations with at least one positive-sale record. It contains only positive-sale days; its `zero_day_interpretation` field warns that missing SKU dates are not automatically zero demand. The complete shop–category panel contains five fields: `shop_id`, `category`, `date`, `units_sold_observed`, and `is_zero_observed_sales`. It has 561,152 rows, 32 shops, 16 categories, 512 series and 1,096 consecutive daily observations per series. The total is 1,580,361 units and 210,972 rows (37.5962%) are zero-sales observations. Aggregating SKU sales to shop–category–date reproduces all 350,180 positive category-day rows exactly; the remaining 210,972 category rows are explicit zero-sales rows in the complete panel.

## 4.4 Inventory and Procurement Reconciliation

The inventory ledger contains 13,272 opening movements, 121,107 purchase movements, 736,683 sale movements, 118,673 expiry-disposal movements and 5,073 damage movements. Every sale movement matches a sales line by order, shop, SKU, lot and quantity. The 121,107 purchase movement groups match received purchase-order/SKU quantities; 1,315 purchase lines have no final received quantity and remain pending at cutoff.

A chronological stock reconstruction identifies 269 negative-balance movement events across 253 shop–SKU combinations. It also identifies 330 positive-sale days with non-positive reconstructed intraday stock. These records are quarantined from SKU exposure labels pending correction. Across the 13,272 inventory-tracked shop–SKU combinations and 1,096 dates, the ledger reconstructs 13,781,575 fully in-stock days and 764,537 days with non-positive intraday balance. These are derived diagnostics, not an independent stock audit. Every zero-sales shop–category day has at least one recorded SKU with positive reconstructed stock during that day; this does not rule out item-level stock-outs or unrecorded assortment.

## 4.5 RQ1 Forecasting Pipeline

For each shop–category series, the feature builder uses only information strictly before the forecast origin. Features include lags of 1, 7, 14, 28, and 56 days; rolling mean, standard deviation, sum, and zero-sales ratio over 7, 14, 28, and 56 days; days since the last positive sale; ADI; positive demand mean and $CV^2$; shop and category identifiers; weekday, day, month, ISO week, and cyclical calendar encodings. No future observed sales are used as predictors.

Four direct cumulative targets are evaluated: 1, 7, 14, and 28 days. Development forecast origins are 1 July, 1 August, and 1 September 2025. The previously inspected fourth-quarter 2025 period is not used for model selection. Historical training examples are sampled every 28 days; each training label window must finish before the corresponding validation origin. A 56-day history is required before a feature row is admitted.

The LightGBM Tweedie candidate uses variance power 1.3, 100 boosting iterations, learning rate 0.035, 31 leaves, minimum child samples 80, row and column subsampling 0.85, subsampling frequency 1, L2 regularisation 2.0 and seed 42. The CatBoost Tweedie challenger uses 120 iterations, depth 6, learning rate 0.04 and L2 regularisation 5. The two-stage CatBoost hurdle uses a 100-iteration occurrence classifier and a 100-iteration positive-quantity regressor, depth 5 and learning rate 0.04; the classifier uses balanced class weights. Its probabilities are not calibrated and its output is not selected for deployment. Croston–SBA and TSB use $\alpha = 0.1$ (and $\beta = 0.1$ for TSB). The weekly seasonal-naive baseline averages up to four previous observations for each matching weekday and sums them across the horizon. All methods are evaluated on the same 512 series at each of the three origins, giving 1,536 series-origin forecasts per horizon.

## 4.6 Software Environment and Reproducibility

The experiment was executed with Python 3.13.5, pandas 2.2.3, NumPy 2.3.5, scikit-learn 1.8.0, and LightGBM 4.6.0. The source panel's SHA-256 checksum is recorded in the accompanying experiment manifest. The implementation writes the per-origin metrics,

aggregate metrics, forecast-level predictions, demand-regime assignments, and regime-stratified metrics as machine-readable CSV files. The raw business dataset is retained in its controlled data location and is not copied into the public source repository.

## 4.7 Current Implementation Boundaries

The executable experiment remains a shop–category observed-sales forecast. It does not establish SKU-level latent demand, price or promotion uplift, or inventory-policy benefit. CatBoost Tweedie and a raw two-stage hurdle have now been evaluated as challengers; the hurdle probabilities remain uncalibrated. The inventory ledger has been reconciled internally, but negative-balance events and positive-sale days with non-positive intraday stock must be quarantined before SKU exposure labels are used. Calibrated prediction intervals, hierarchical reconciliation, point-in-time price and promotion features, and inventory replay remain subsequent tasks. The package's provenance conflict also prevents real-world effectiveness claims.

# Chapter 5

# Experimental Results

## 5.1 Evidence Boundary and Panel Integrity

The current Drive manifest says the package is real, while the 2 November repository onboarding record classifies a same-seed, same-count package as synthetic. The current accessible SKU sheet does not expose the previously reported `synthetic_flag` column. This conflict remains unresolved. All numerical results in this chapter are therefore reproducible controlled development results on the supplied package, not verified real-world SME accuracy. The previously inspected October–December 2025 period is excluded from model selection and is not an untouched final test.

The shop–category panel contains 561,152 observations, 512 unique series and 1,096 daily observations per series. It totals 1,580,361 units; 37.5962% of rows are explicit zero-sales observations. There are no duplicate shop–category–date keys, missing required values, zero-flag mismatches or missing dates within the 512 series. Transaction and inventory reconciliation results are reported in Chapter 4.

## 5.2 Demand-Regime Distribution

Regimes are calculated using data available through 30 June 2025, before the first development origin. The ADI and $CV^2$ thresholds are 1.32 and 0.49. The panel contains 342 lumpy series, 153 erratic series, 13 smooth series and only 4 intermittent series.

Table 5.1: Historical demand-regime distribution across 512 shop–category series.

| **Regime** | **Series** | **Share** | **Mean zero-sales ratio** |
|---|---|---|---|
| Lumpy | 342 | 66.80% | 47.62% |
| Erratic | 153 | 29.88% | 15.65% |
| Smooth | 13 | 2.54% | 3.64% |
| Intermittent | 4 | 0.78% | 83.94% |
| Total | 512 | 100.00% | 37.60% |

Four intermittent series are not enough independent support for regime-specific automatic model switching. Repeated forecasts of the same four series do not create additional independent businesses.

## 5.3 Multi-Model Forecasting Performance

The fixed-origin experiment uses 1 July, 1 August and 1 September 2025. Each of the 512 series is forecast at each origin, giving 1,536 series-origin observations per horizon. The six candidates are LightGBM Tweedie, CatBoost Tweedie, a raw two-stage CatBoost hurdle, weekly seasonal-naive, Croston–SBA and TSB. All models use the same target definition and historical information at each origin.

TABLE 5.2: Pooled development WAPE by forecast horizon and candidate model.

| Horizon | LightGBM WAPE | CatBoost WAPE | Hurdle WAPE | Croston–SBA WAPE | TSB WAPE | Seasonal naive |
|---|---|---|---|---|---|---|
| 1 day | 79.80% | 79.98% | **79.23%** | 84.01% | 84.78% | 92.21% |
| 7 days | **33.74%** | 34.20% | 35.26% | 37.74% | 38.62% | 40.77% |
| 14 days | **24.11%** | 24.78% | 24.54% | 30.27% | 32.10% | 33.97% |
| 28 days | **18.57%** | 19.28% | 18.84% | 25.62% | 27.38% | 29.16% |

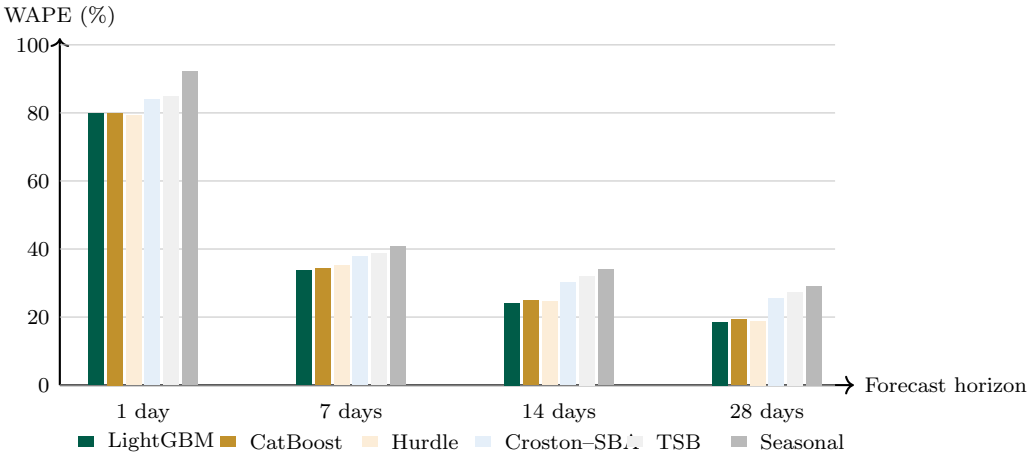



FIGURE 5.1: Measured development WAPE for the six candidates at four direct cumulative horizons. The one-day task remains difficult; lower WAPE at longer horizons reflects a different cumulative target and must not be interpreted as equivalent daily accuracy.

LightGBM has the lowest pooled WAPE at seven, fourteen and twenty-eight days. The raw hurdle candidate has the lowest one-day WAPE, but its one-day aggregate bias is −28.18%, compared with −21.69% for LightGBM. The hurdle probabilities are uncalibrated. Its lower one-day WAPE therefore does not justify promotion without calibration and a separate cost-sensitive analysis of under-forecasting.

Table 5.3: Forecast bias and horizon-scaled weekly MASE for the principal candidates.

| Horizon | LightGBM bias | CatBoost bias | Hurdle bias | LightGBM MASE |
|---|---|---|---|---|
| 1 day | −21.69% | −21.24% | −28.18% | 0.793 |
| 7 days | −6.59% | −7.64% | −12.80% | 0.326 |
| 14 days | −2.42% | −2.86% | −2.86% | 0.226 |
| 28 days | −2.10% | −2.84% | −1.01% | 0.175 |

Horizon-scaled weekly MASE divides absolute forecast error by the series-specific mean absolute lag-7 difference multiplied by the forecast horizon. It is a scale-normalised diagnostic, not a direct business-cost measure. The one-day WAPE remains high; longer-horizon results must not be interpreted as evidence of equally accurate daily replenishment forecasts.

## 5.4 Seven-Day Stability and Shop-Level Heterogeneity

LightGBM has the lowest seven-day pooled WAPE at each development origin: 32.52% on 1 July, 33.86% on 1 August and 34.92% on 1 September. CatBoost records 32.71%, 34.26% and 35.72%; the raw hurdle records 33.81%, 35.54% and 36.52%.

Table 5.4: Seven-day WAPE by development origin.

| Origin | LightGBM | CatBoost | Hurdle | Croston–SBA | TSB |
|---|---|---|---|---|---|
| 1 July 2025 | **32.52%** | 32.71% | 33.81% | 37.91% | 39.82% |
| 1 August 2025 | **33.86%** | 34.26% | 35.54% | 36.14% | 35.82% |
| 1 September 2025 | **34.92%** | 35.72% | 36.52% | 39.21% | 40.24% |

Across shops, LightGBM has lower seven-day WAPE than CatBoost in 22 of 32 shops; across categories it wins in 14 of 16. Median shop-level WAPE is 35.89% for LightGBM and 36.79% for CatBoost. The model is not uniformly best for every shop, so per-shop monitoring and abstention remain necessary.

Table 5.5: Cluster-bootstrap seven-day WAPE differences.

| **Challenger** | **Median difference** | **95% interval** | **Interpretation** |
|---|---|---|---|
| CatBoost Tweedie | −0.45 pp | [−0.85, −0.08] pp | LightGBM lower WAPE |
| Raw CatBoost hurdle | −1.51 pp | [−2.12, −0.93] pp | LightGBM lower WAPE |
| Weekly seasonal-naive | −7.01 pp | [−8.55, −5.59] pp | LightGBM lower WAPE |
| Croston–SBA | −3.97 pp | [−5.04, −2.94] pp | LightGBM lower WAPE |
| TSB | −4.90 pp | [−6.11, −3.72] pp | LightGBM lower WAPE |

The intervals resample 512 shop–category series with 2,000 bootstrap replicates. They quantify variation across the development series under the fixed-origin protocol; they are not uncertainty intervals for future real-world SME performance and do not replace an untouched test.

## 5.5 Performance by Demand Regime

At seven days, CatBoost has slightly lower WAPE than LightGBM in the erratic group (28.50% versus 28.67%), while LightGBM performs better in the lumpy and smooth groups. Croston–SBA has lower WAPE in the intermittent group (77.02% versus 106.30% for LightGBM), but that subset contains only four series, twelve series-origin forecasts and 25 actual units. The raw hurdle has WAPE 83.18% in the same subset but under-forecasts total demand by 74.62%. None of these sparse subgroup results justifies automatic model routing.

Table 5.6: Seven-day WAPE by historical demand regime.

| **Regime** | **Cases** | **LightGBM** | **CatBoost** | **Hurdle** | **Croston–SBA** |
|---|---|---|---|---|---|
| Erratic | 459 | 28.67% | **28.50%** | 28.57% | 32.38% |
| Lumpy | 1,026 | **42.06%** | 42.85% | 45.60% | 47.09% |
| Smooth | 39 | **19.89%** | 22.03% | 20.60% | 20.69% |
| Intermittent | 12 | 106.30% | 121.62% | 83.18% | **77.02%** |

## 5.6 Model and Solution Design Decision

The development evidence supports a conservative two-layer forecasting design. The forecast layer retains LightGBM Tweedie as the fallback candidate for shop–category horizons of seven days and longer; CatBoost Tweedie remains a challenger; the hurdle model remains experimental until its occurrence probabilities are calibrated and its bias

is controlled; Croston–SBA and TSB remain specialist candidates for intermittent series. The selection layer compares candidates on identical origins, horizons and target rows, and selects a specialist only when at least three origins and at least twenty independent series per regime per origin are available. Otherwise the pooled fallback is kept.

The next SKU-level design is availability-aware. Missing SKU dates become zero only when the SKU is active and positive stock is established throughout the day. Negative-ledger events, stock-out days, unknown assortment and closed-shop days are censored. The simulated unmet-demand table is not used as observed target data. Calibrated intervals and hierarchical reconciliation follow only after the eligible panel and aggregation identities are frozen. RQ3 inventory-policy replay and RQ4 owner-facing explanation evaluation remain separate experiments with their own evidence requirements.

## 5.7 Threats to Validity and Summary

The current result is bounded to a supplied shop–category panel, three development origins and four cumulative horizons. Its provenance is unresolved, and the package cannot currently support a claim of verified real-world SME forecasting accuracy. The previously discussed 83.98%, 84.04% and 84.56% WAPE values are not compared with this experiment because their exact source code, target, aggregation, forecast origins and metric denominator have not been reproduced on identical rows. The Q4-2025 period is not an untouched final test.

The evidence supports an engineering decision for the next development cycle: keep a pooled fallback, add specialist candidates only behind validation and support gates, correct the SKU target construction using inventory exposure, calibrate uncertainty and probabilities, and evaluate inventory decisions separately from forecast error. No model is declared a universal winner and no accuracy target is forced.

# Chapter 6

# Conclusions

## 6.1 Research Summary

This progress report establishes a data-audited, multi-candidate RQ1 development workflow for a proposed SME business analytics framework. The current Drive manifest and the repository's historical onboarding record conflict on whether the supplied package is real or synthetic. The evidence class is therefore provenance-conflict, and the reported scores are controlled development results rather than verified real-world Bangladeshi SME accuracy.

The source tables reconcile internally: all sales lines join to orders, the prepared SKU daily table is reproduced from transaction lines, the positive category-day rows are reproduced from SKU aggregation, and sale inventory movements match sales lines. The inventory audit nevertheless finds 269 negative-balance events and 330 positive-sale days with non-positive reconstructed intraday stock. These anomalies must be quarantined before SKU-level availability labels are accepted.

In the fixed-origin shop–category experiment, LightGBM Tweedie achieved the lowest pooled WAPE at 7-, 14- and 28-day horizons. CatBoost remains a challenger. The raw hurdle candidate achieved a lower one-day WAPE but under-forecast materially and used uncalibrated occurrence probabilities. Only four of 512 series were classified as intermittent; this is insufficient support for automatic regime-based model routing. The one-day WAPE remains high, and the results do not establish SKU-level or unconstrained-demand accuracy.

The resulting architecture is evidence-gated: source and availability validation precede target construction; pooled and specialist candidates are compared on identical rolling origins; specialist routing requires adequate independent-series support; calibrated uncertainty and hierarchical reconciliation follow; and inventory actions are evaluated under actual business constraints. The contribution is the reproducible and auditable process, not a claim that a newly named model has already solved SME forecasting.

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